SFU NatLangLab

CMPT 413/713: Natural Language Processing

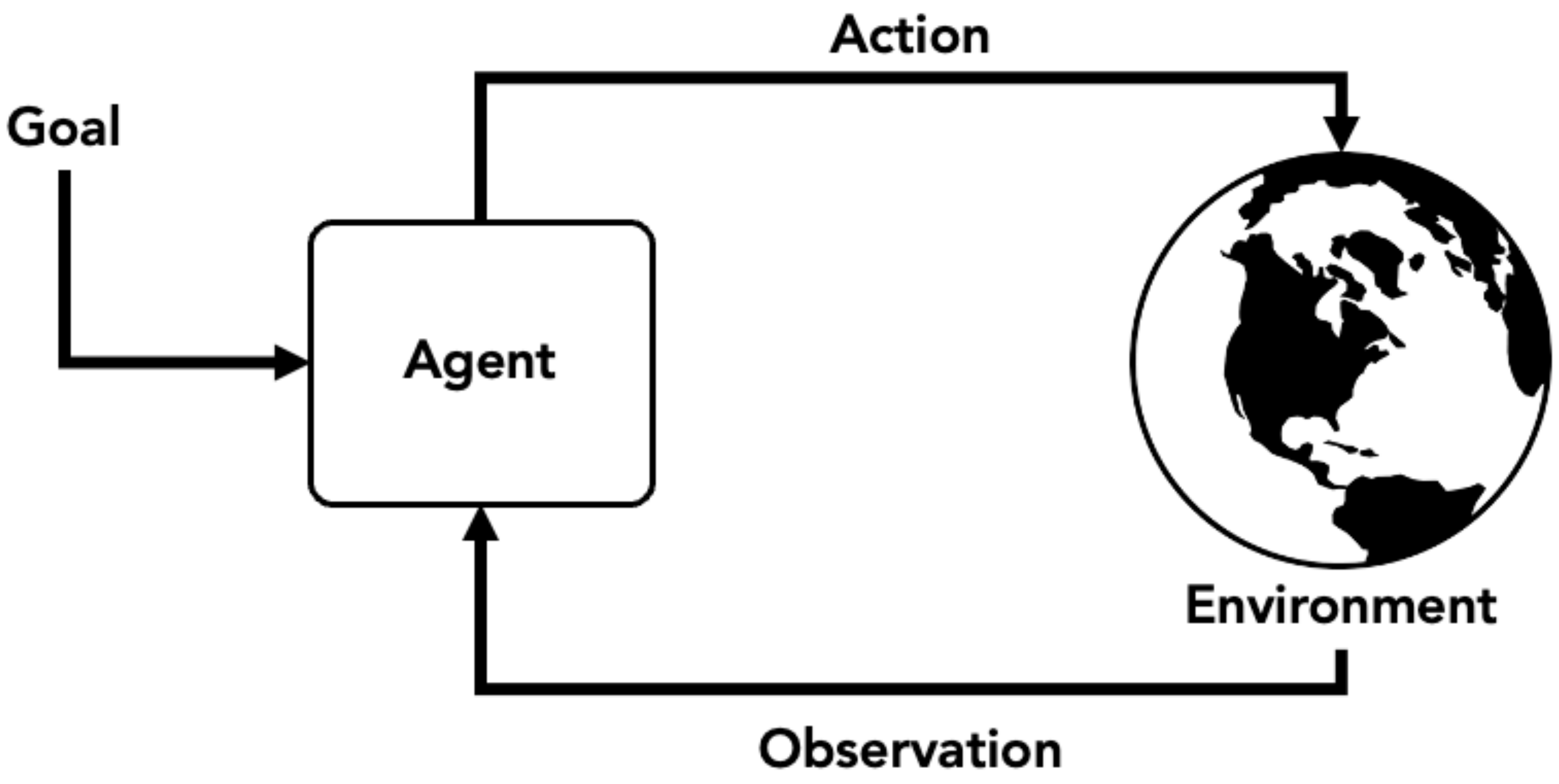
LLM agents

Spring 2024

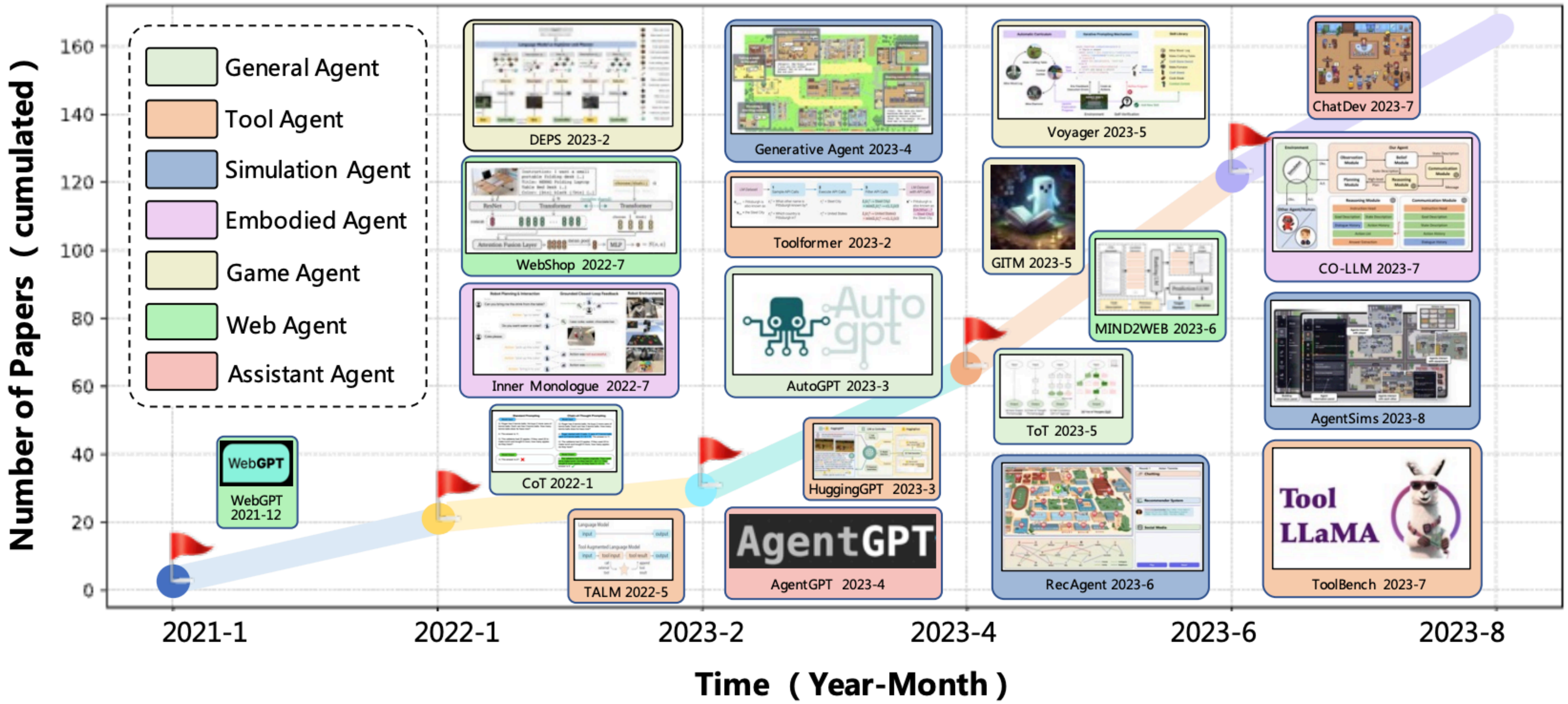
2024-03-27

LLMs as agents

- There is a lot of knowledge in LLMs
- But they can't "act"
- Can we leverage LLMs to build a "smart" agent that can interact with the environment to achieve given goals?



Lots of work!



A Survey on Large Language Model based Autonomous Agents [Wang et al, 2023]

Learn about LLM agents from Berkeley!

Advanced LLM Agents MOOC ▶ Play all

https://rdi.berkeley.edu/adv-llm-agents/sp25

CS 194/294-280 (Advanced LLM Agents) - Lecture 6,...
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LLM Agents MOOC ▶ Play all

Large Language Model Agents MOOC F24

https://llmagents-learning.org/f24

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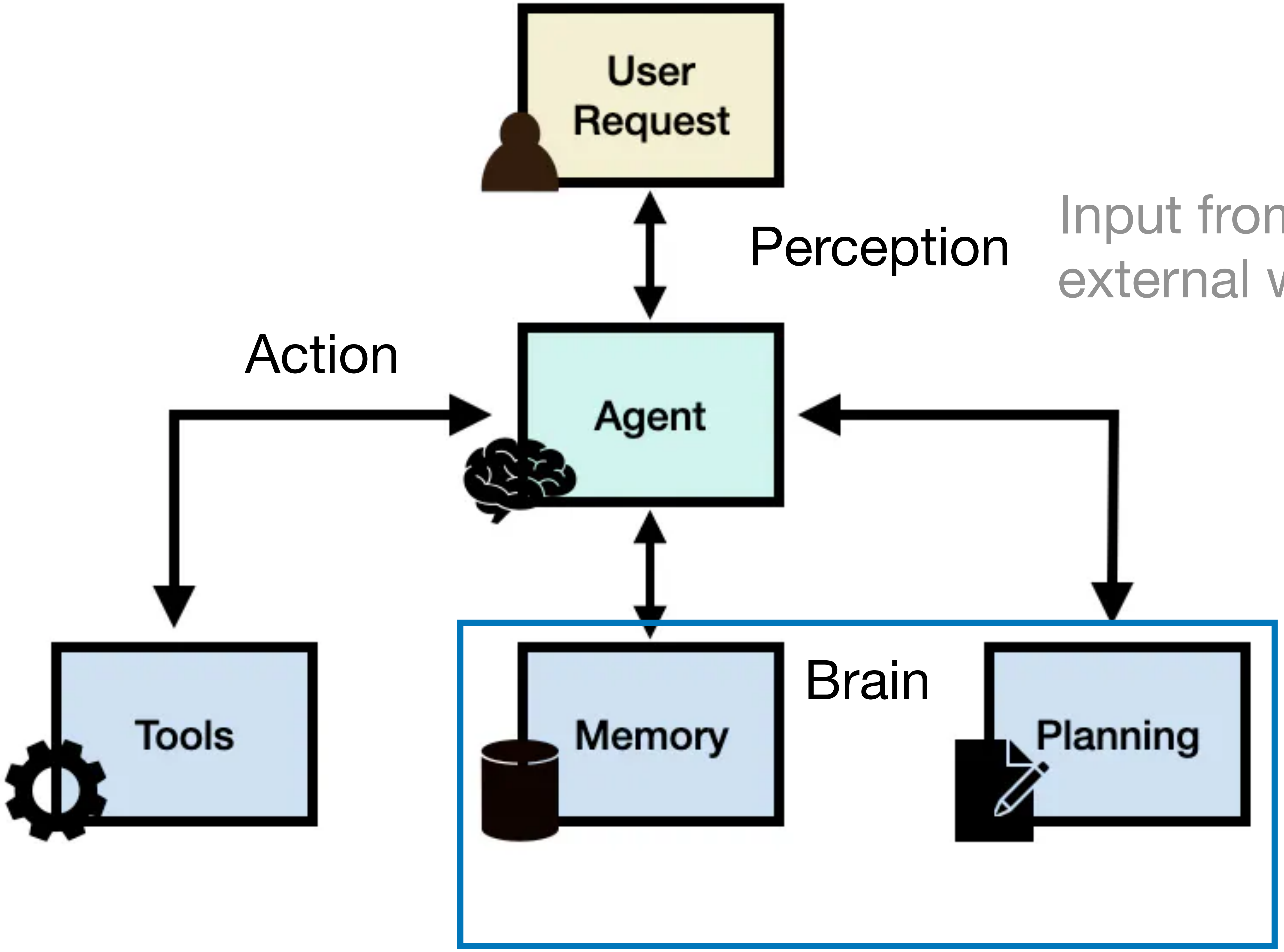
LLM Agent Framework

API calls to external services, expert modules, etc

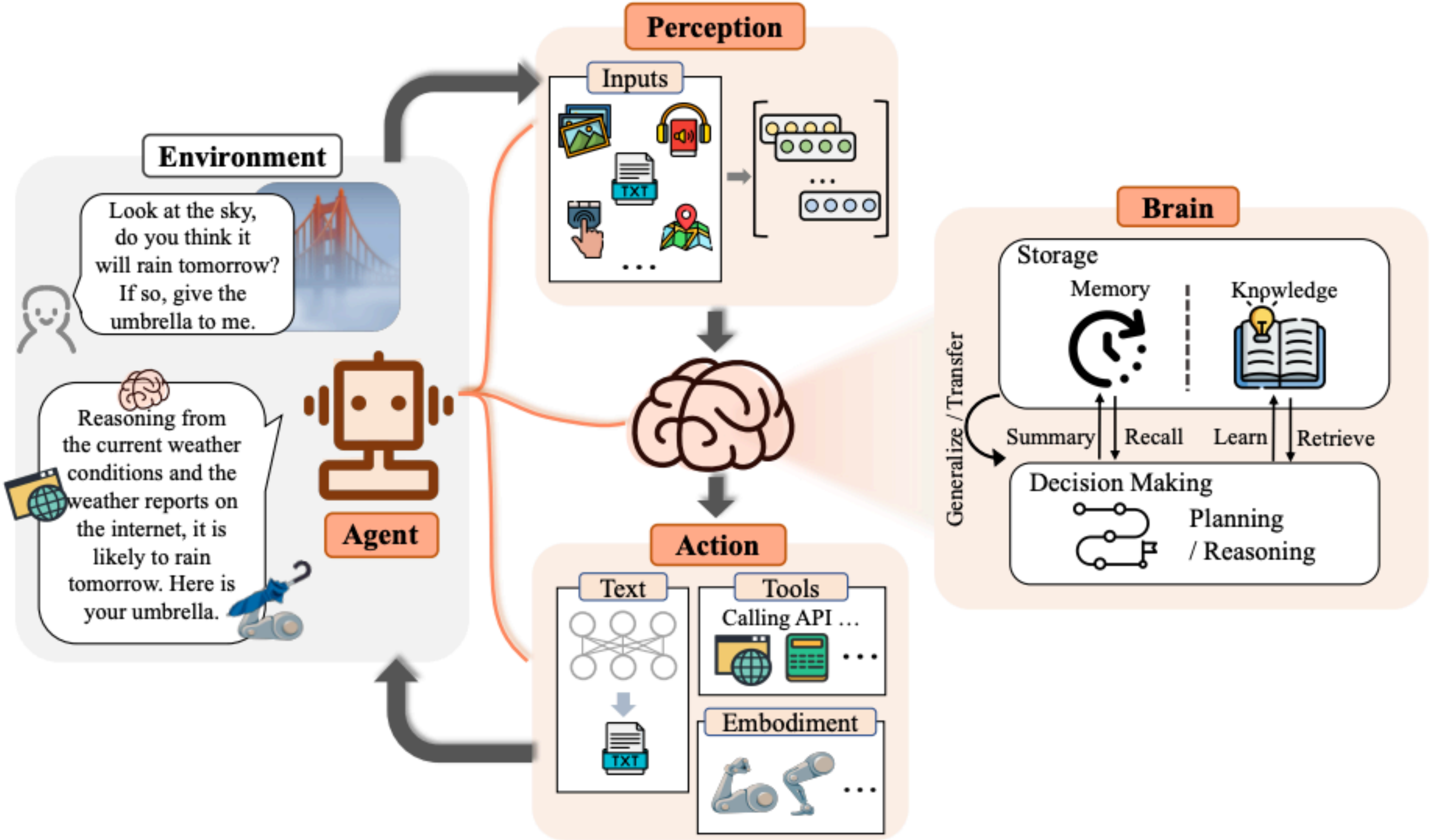
Calendar()
Calculator()
ExecuteCode()
Search()



Input from the external world

Reflection
Self-critics
Chain-of-thoughts
Subgoal decomposition

https://www.promptingguide.ai/research/llm-agents



The Rise and Potential of Large Language Model Based Agents: A Survey [Xi et al, 2023]

Planning

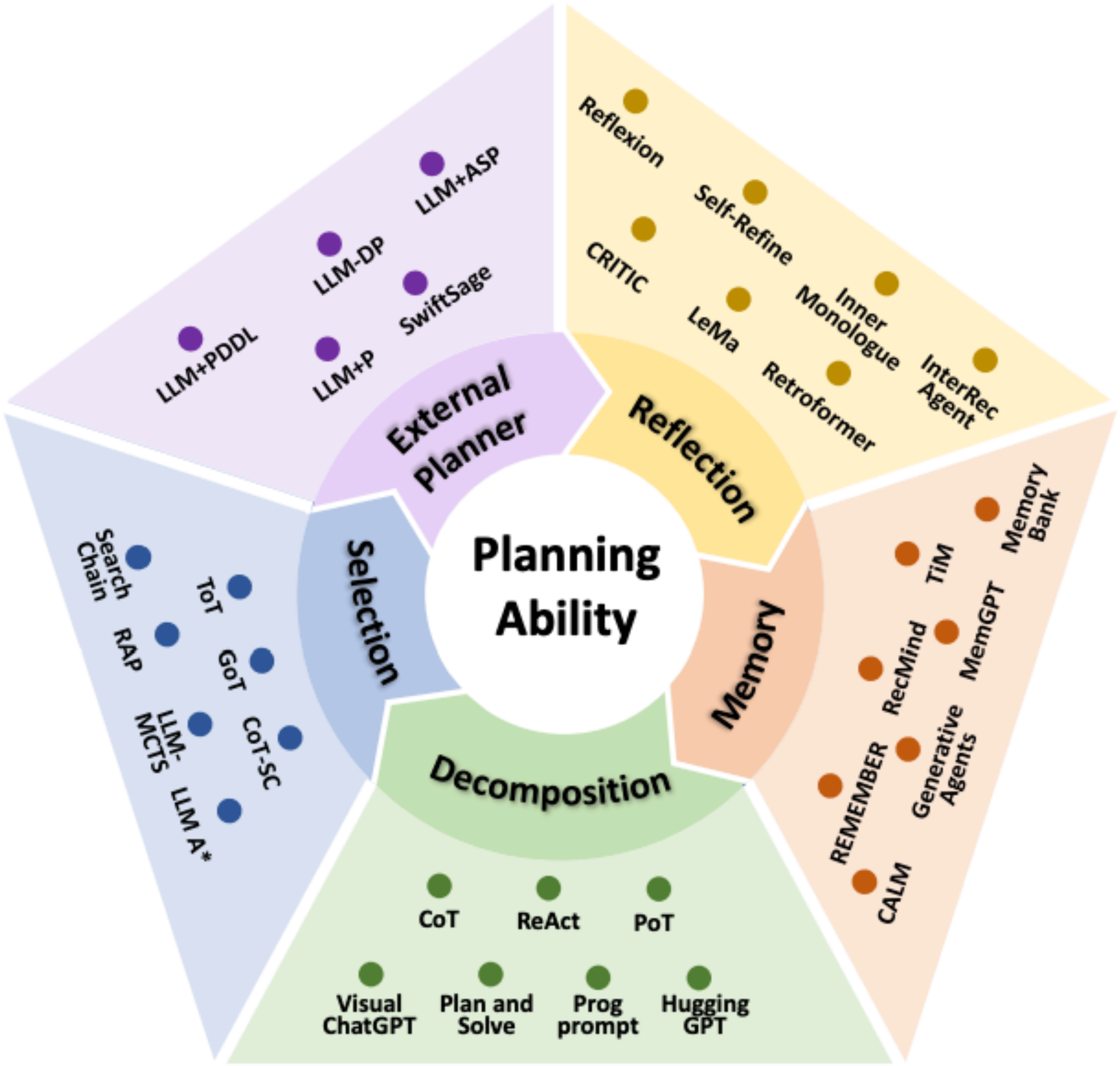
Planning

- What does the agent need to do to accomplish a specified goal?
- Low-level vs high-level planning
  - High-level plan: Identify subgoals for a long-horizon task
  - Low-level plan (sometime low-level control): Identify sequence of actions
- Traditionally use symbolic reasoning
  - Hard to recover from errors
  - Difficult to convert expert knowledge into planning languages such as PDDL (Planning Domain Definition Language)
- Use of LLM for planning
  - Lot of expert domain knowledge already encoded in language
  - Can we use LLMs to help us plan?

Taxonomy for planning with LLMs

Given environment and sequence of actions, and an overall task goal, identify subgoals and actions

- **Task decomposition** - figure out subgoals, do planning for subgoals if needed
- **Multi-plan selection** - generate multiple plans and then select one
- **External planner** - LLM used to formalized the problem which is passed to an external planner
- **Reflection and refinement** - After obtaining a plan, the LLM future reflects on the plan and refine it to fix any issues with the original plan
- **Memory-augmented planning** - Uses external memory to retrieve information (common sense knowledge, domain-specific knowledge, etc) and then determines plan based on that



Understanding the planning of LLM agents: A survey [Huang et al, 2024]

Self-Reflection

(1) Hotspot QA

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple … originally designed to control the Front Row media center program …

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports',' Front Row (software) ', …]

Thought 3: Front Row is not found. I need to search Front Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

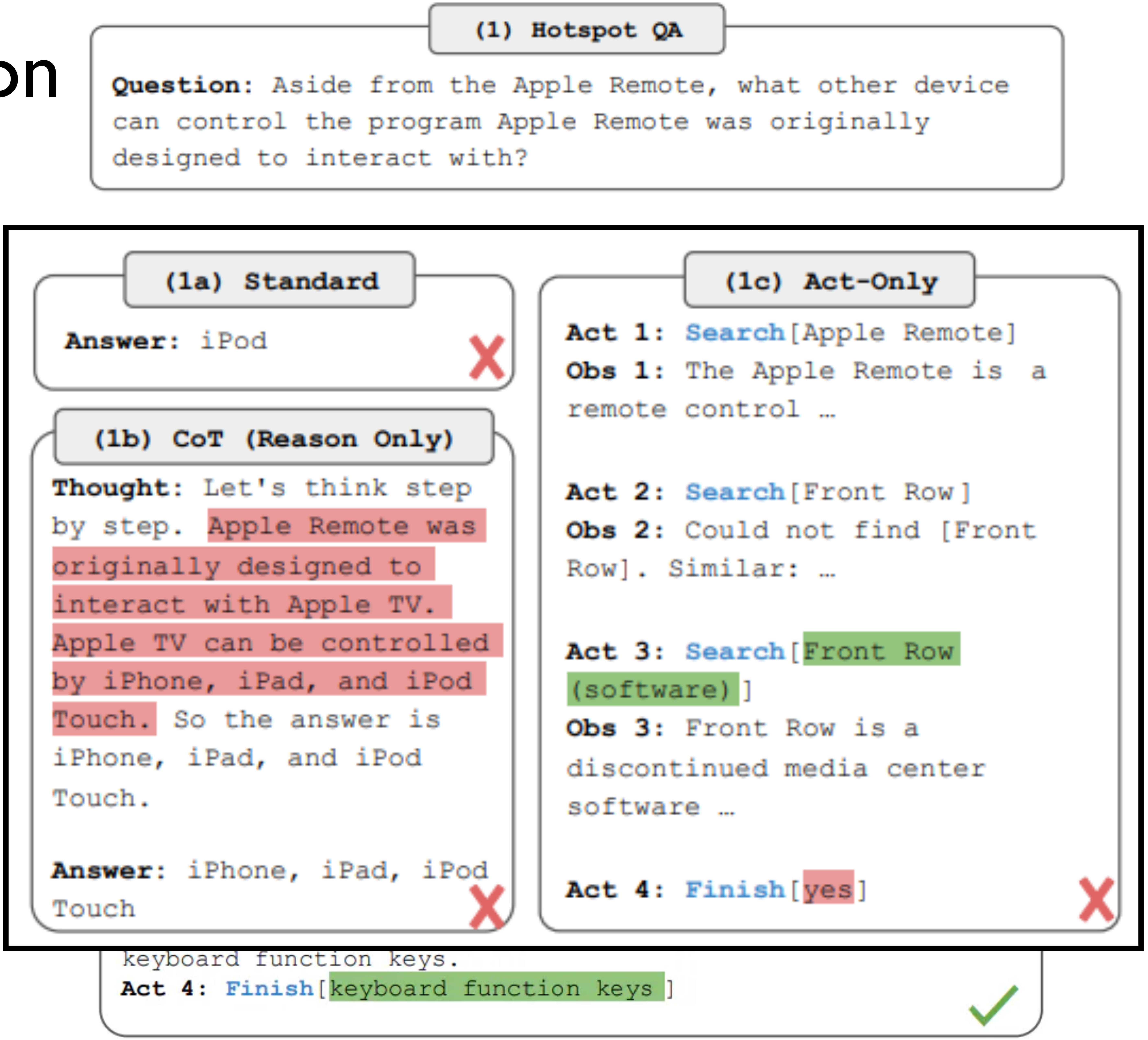
Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4: Finish[keyboard function keys]

- ReAct: interleave thought, action, and observation
- Prompting to combine reasoning with actions

ReAct: Synergizing Reasoning and Acting in Language Models [Yao et al. 2022]

Self-Reflection

- Comparison with other ways of prompting



[ReAct: Synergizing Reasoning and Acting in Language Models](#) [Yao et al. 2022]

Memory

Types of memory

Memory

- Sensory memory
  few seconds
  - Iconic memory (visual)
  - Echoic memory (auditory)
  - Haptic memory (touch)

  Learn embeddings for raw inputs

- In-context learning
  Short-term memory (Working memory)
  20-30 seconds

- Long-term memory
  External vector store (fast retrieval for access)
  - Explicit / Declarative memory (conscious)
    Recalling events and facts
    - Episodic memory (life events)
    - Semantic memory (facts, concepts)
  - Implicit / Procedural memory (unconscious; skills)
    Example: Riding a bike

https://lilianweng.github.io/posts/2023-06-23-agent/

Different types of ANNs

Vector transformation and encoding (focus on reduced memory)

- Quantization (FAISS, ScaNN)

Structures for searching (focus on fast search)

- Hashes (LSH)
- Trees (ANNOY)
- Graphs (HNSW)

Methods can be combined

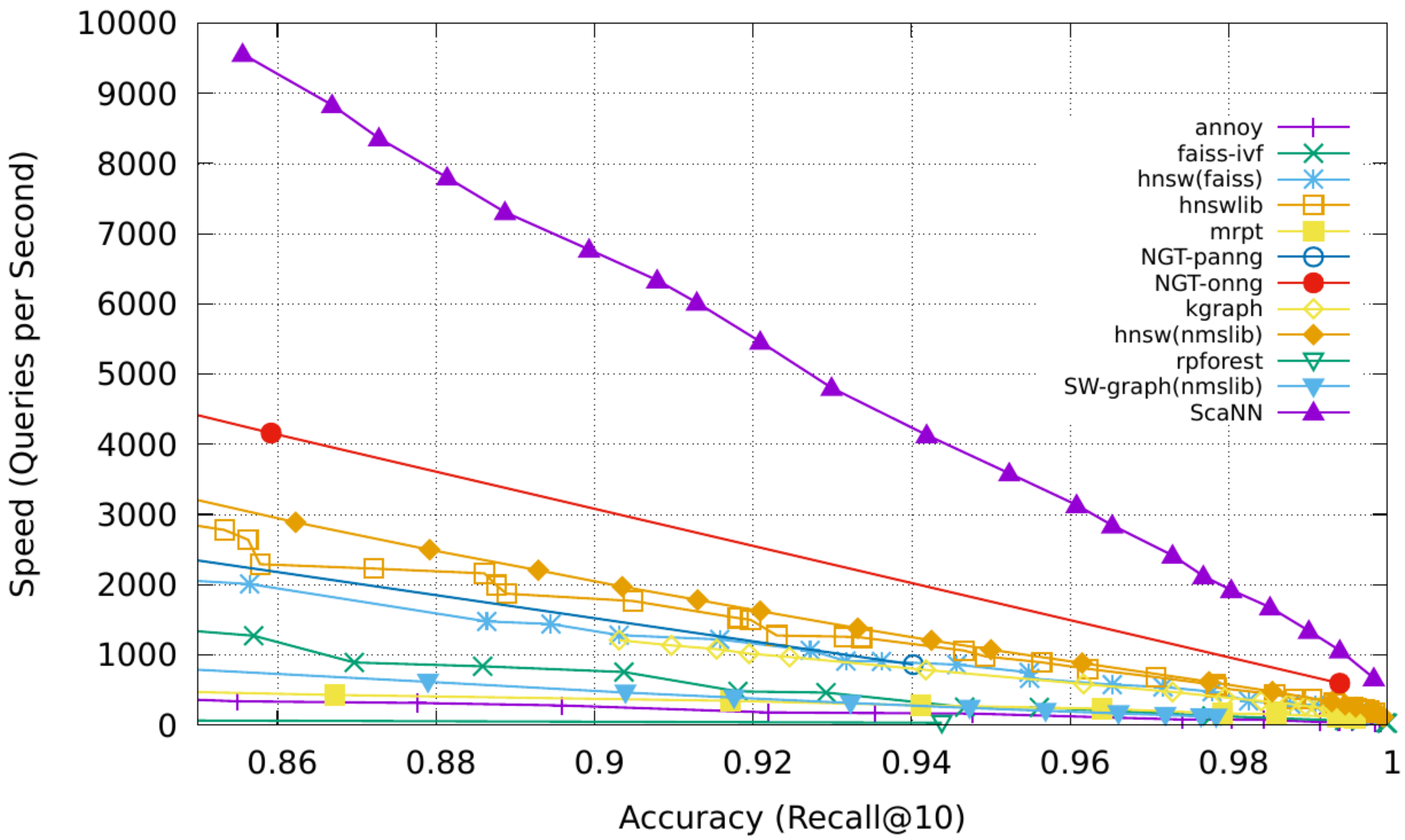
FAISS library: https://github.com/facebookresearch/faiss

Efficient retrieval from memory

Approximate nearest neighbours (ANN) algorithms to return top k nearest neighbours using maximum inner product search (MIPS)

- LSH (Locality Sensitive Hashing) - hashing function so that similar inputs are mapped to same buckets with high probability
- ANNOY (ANN Oh Yeah) - Random projection trees where nodes splits input space into half
- HNSW (Hierarchical Navigable Small World)
  - Hierarchical layers of small-world graphs (points in the bottom layers)
  - Can be used with FAISS
- FAISS (Facebook AI Similarity Search) - vector quantization - partition vector space into clusters
- ScaNN (Scalable Nearest Neighbours) - Anisotropic vector quantization (quantize points while maintaining distances)

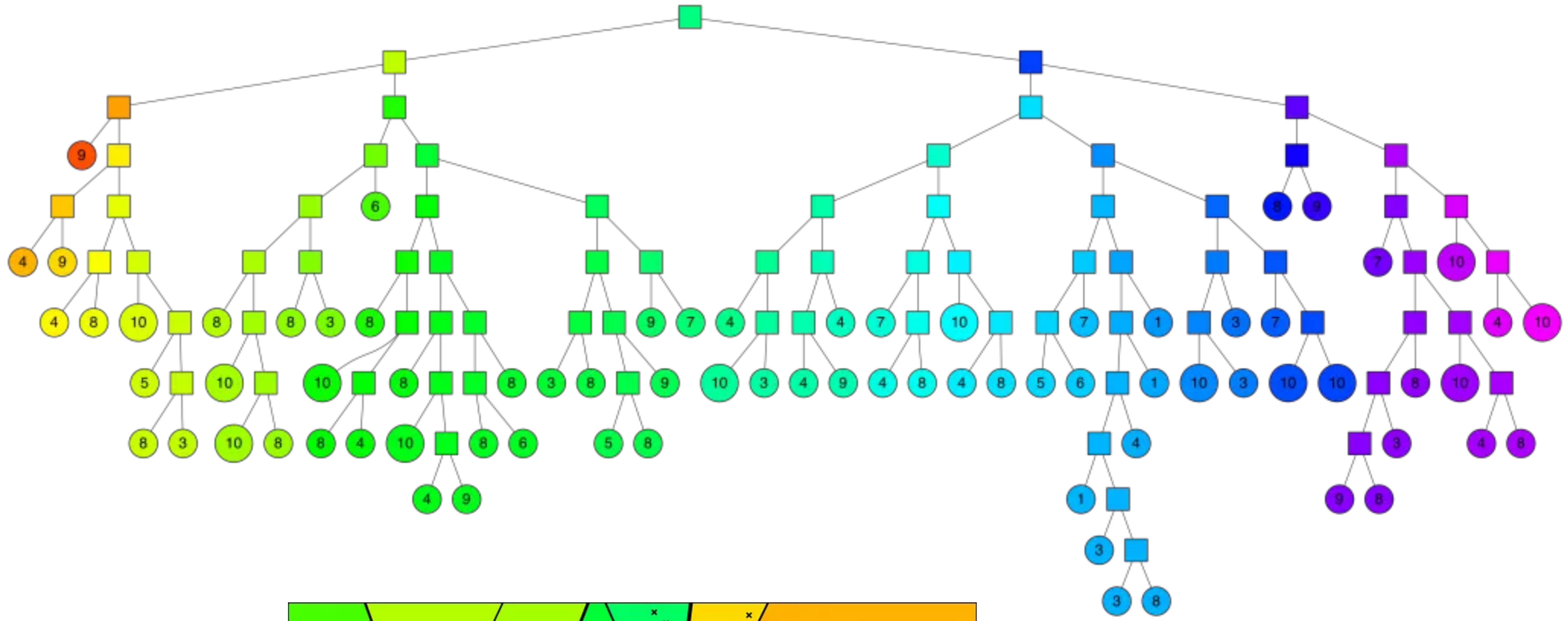
https://lilianweng.github.io/posts/2023-06-23-agent/



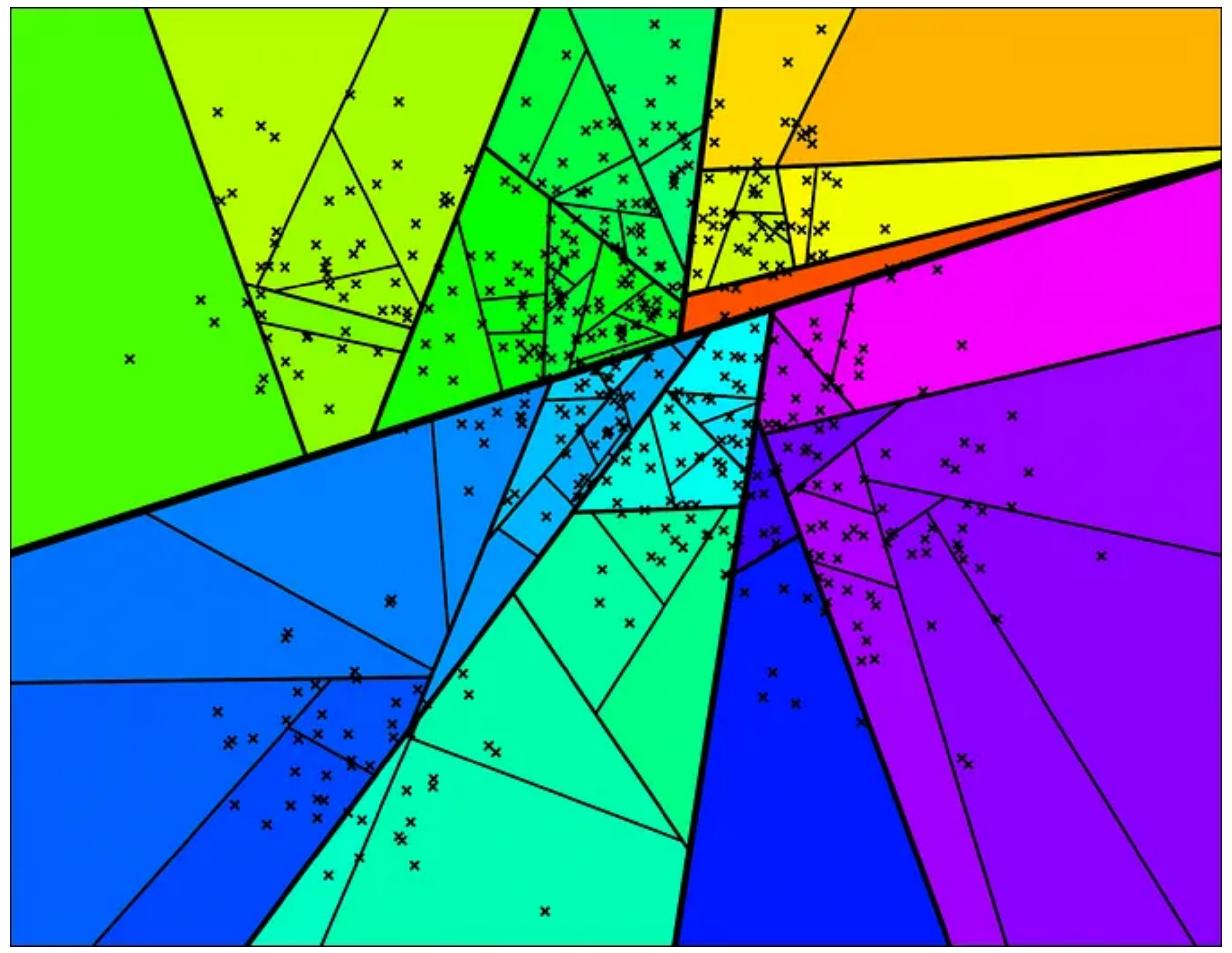
https://blog.research.google/2020/07/announcing-scann-efficient-vector.html

Note: nmslib focus is on non-metric spaces

ANNOY



- Tree-based ANN
- Construction
  - Pick two points at random and split the space in two by the hyperplane between the two points
  - Repeat recursively until number of points associated with a node is of size < threshold
  - Parameters: number of trees
- Search



https://erikbern.com/2015/10/01/nearest-neighbors-and-vector-models-part-2-how-to-search-in-high-dimensional-spaces.html

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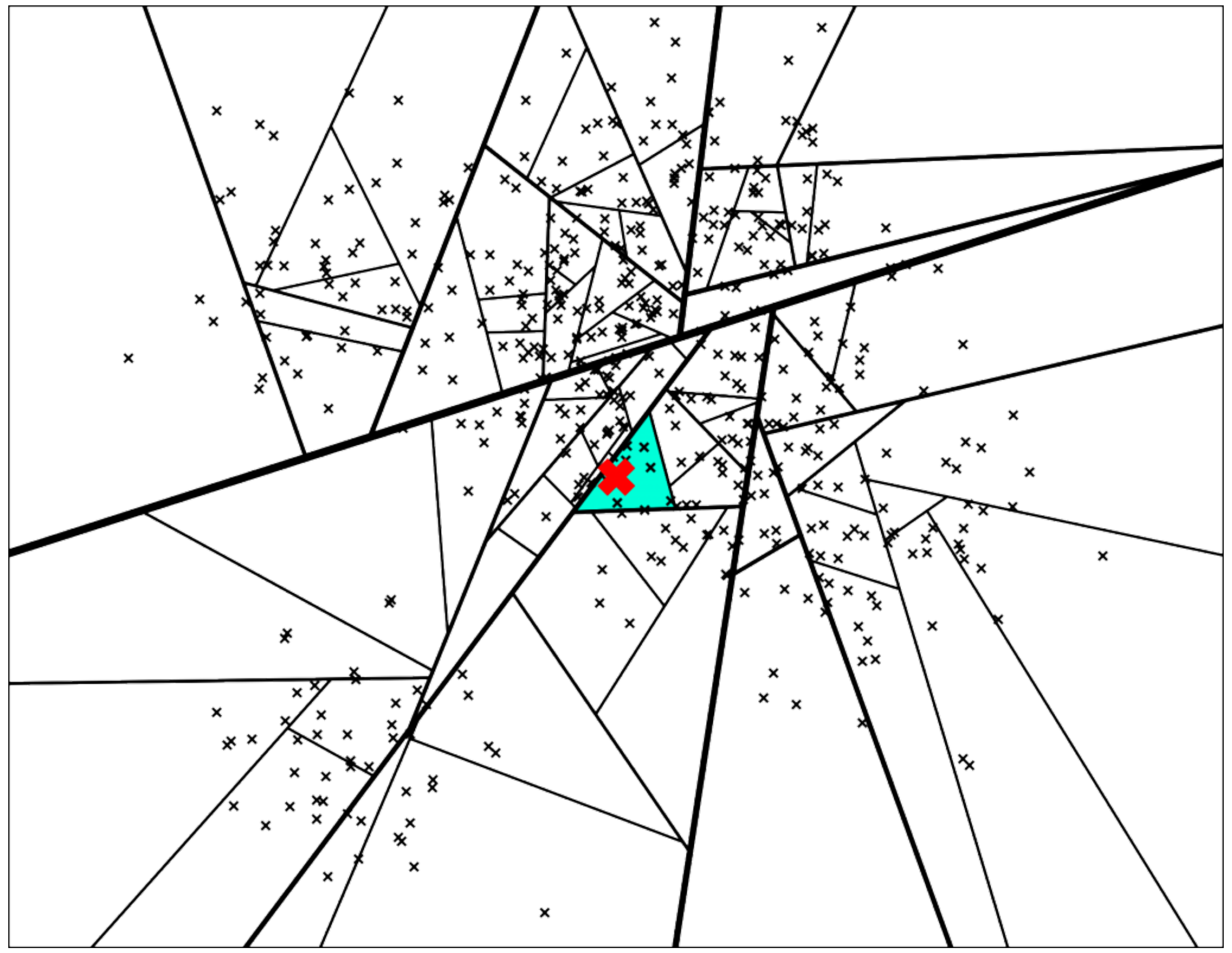
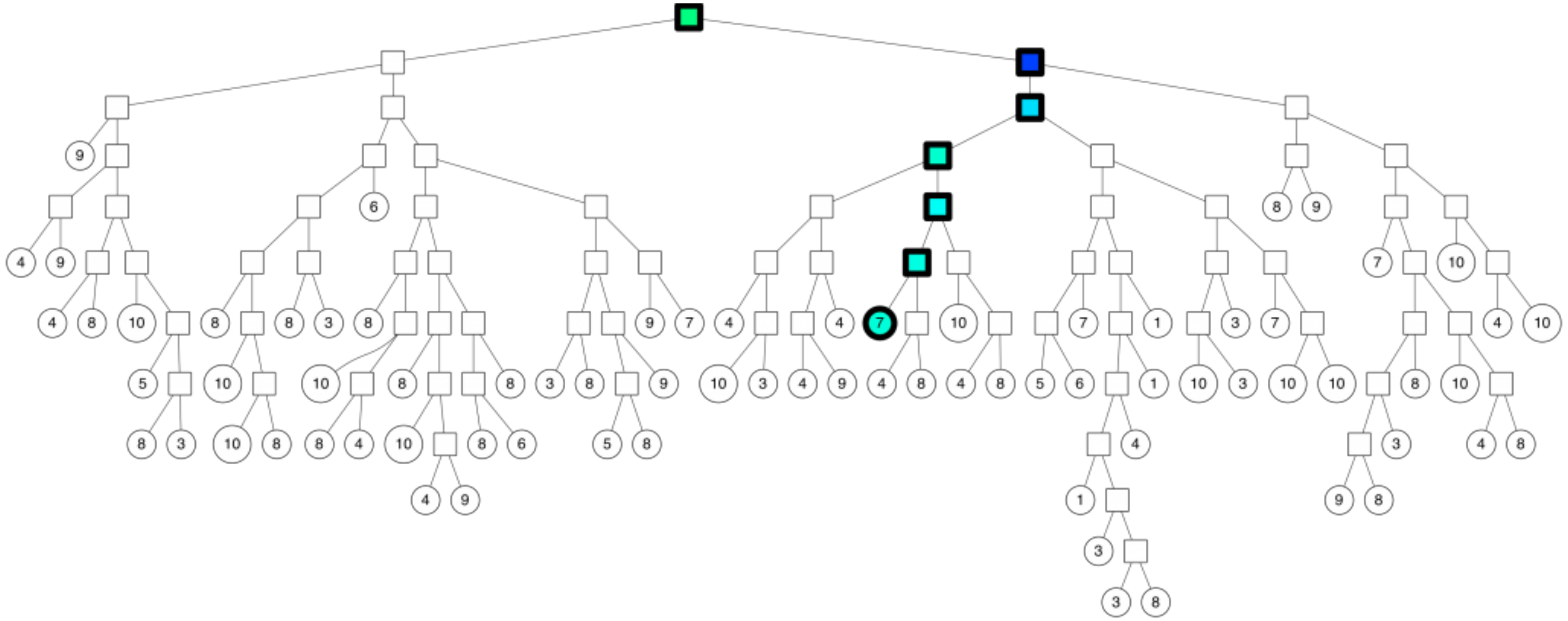


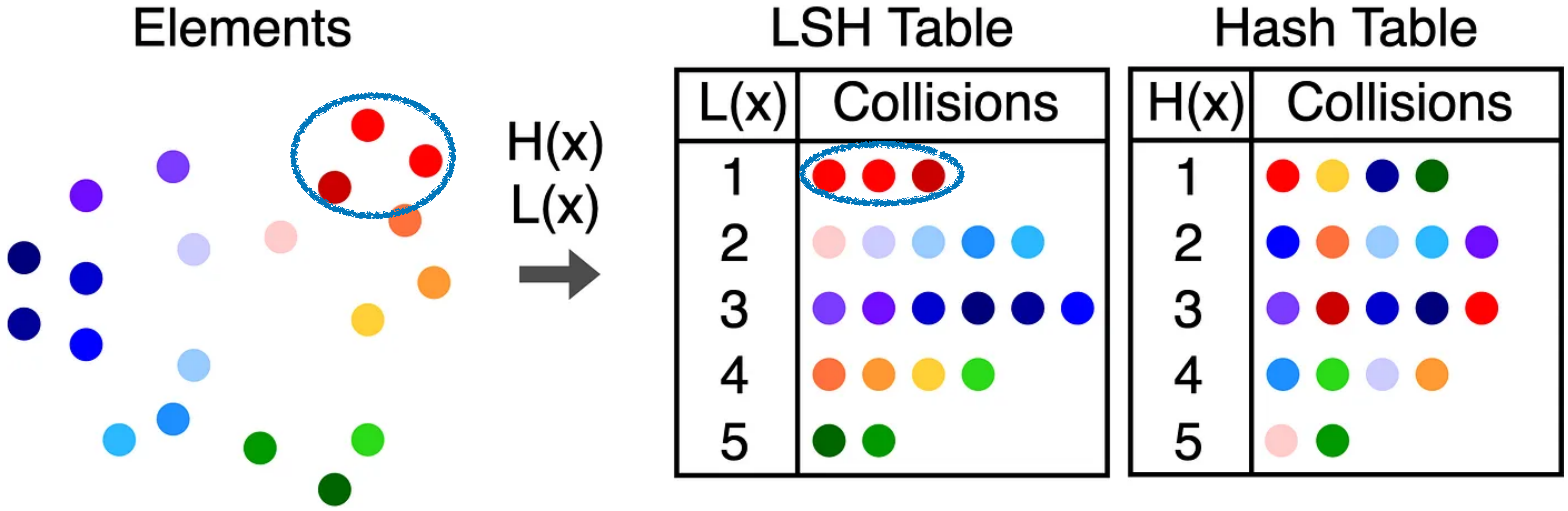
- Tree-based ANN
- Construction
  - Pick two points at random and split the space in two by the hyperplane between the two points
  - Repeat recursively until number of points associated with a node is of size < threshold
  - Parameters: number of trees
- Search
  - Traverse down the tree from the root



https://erikbern.com/2015/10/01/nearest-neighbors-and-vector-models-part-2-how-to-search-in-high-dimensional-spaces.html

Locality Sensitive Hashing (LSH)

- Hash-based ANN
- Construction
  - Map data points into buckets so points near each other are in the same buckets with high probability
- Search



Hierarchical navigable small world (HNSW)

- Graph based ANN
- Hierarchical layers of Navigable Small World Graphs

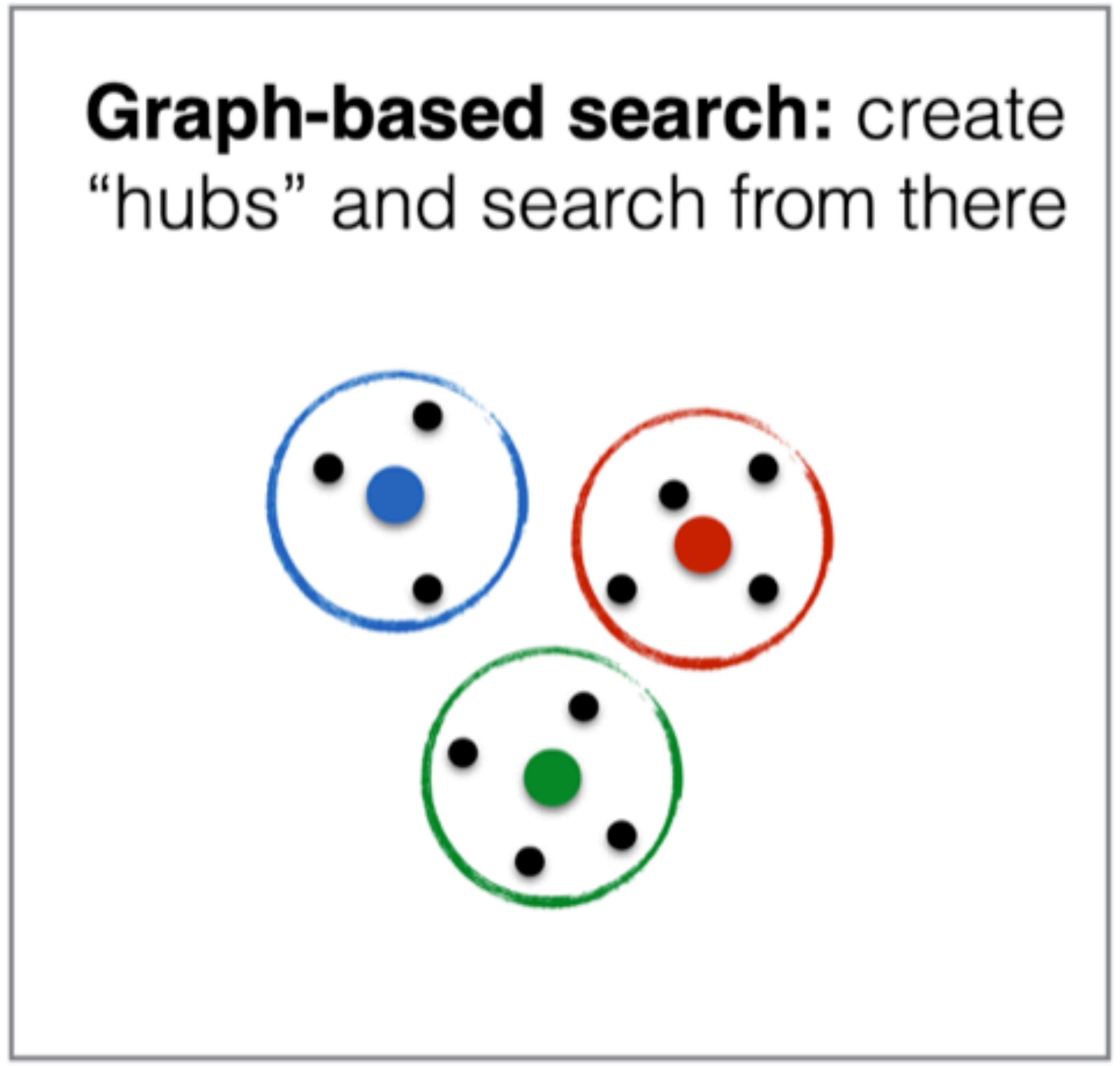
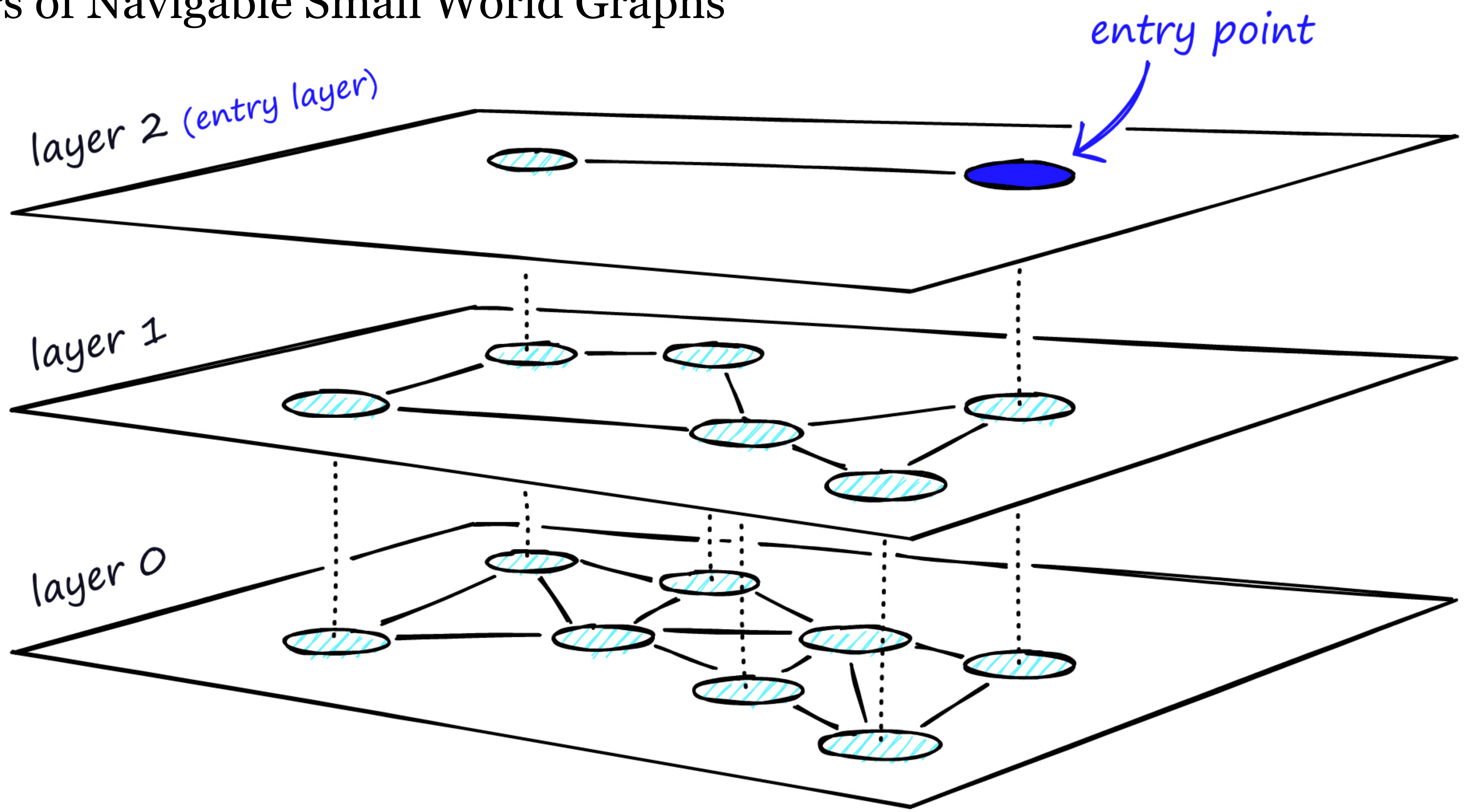


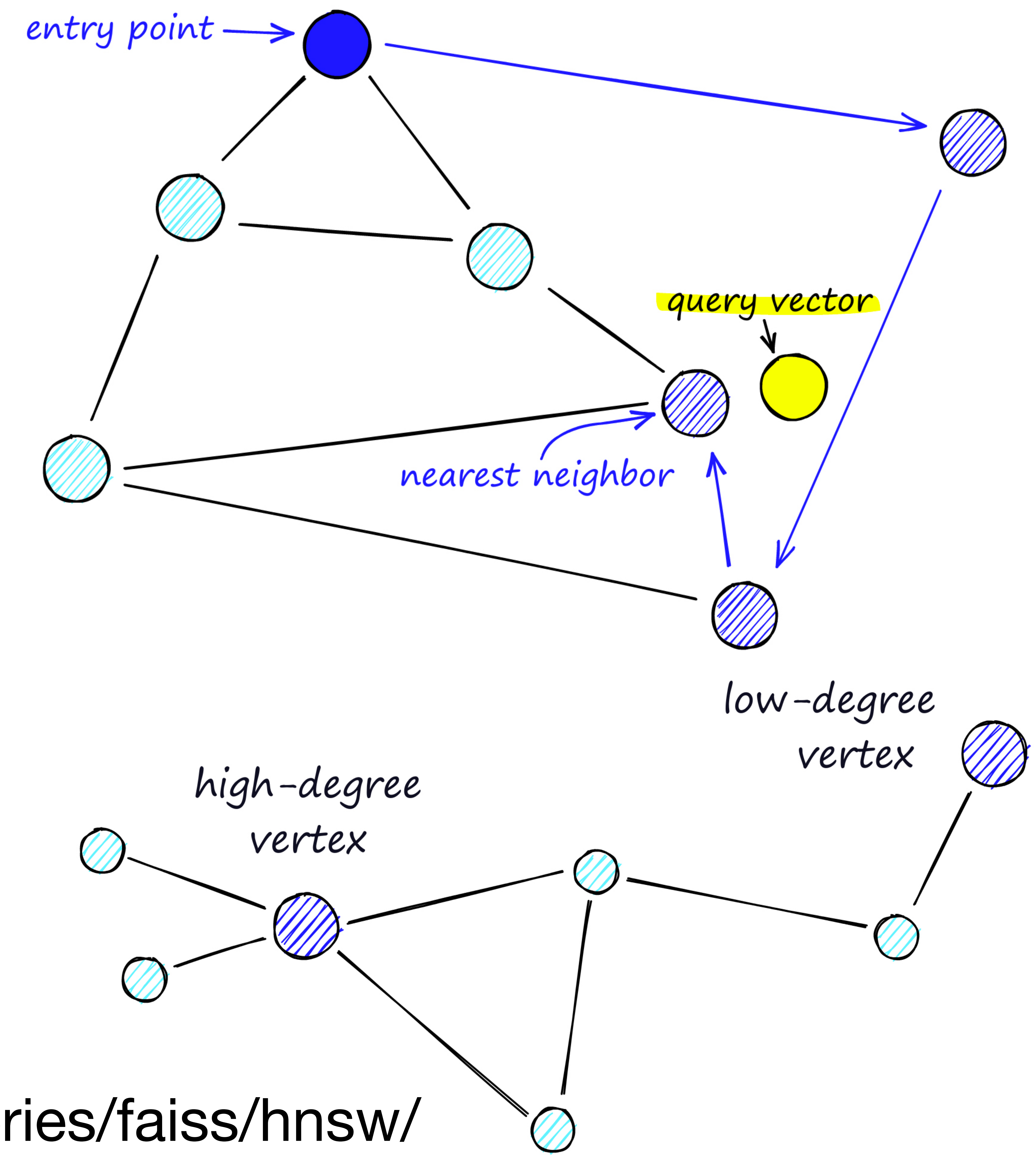
Figure credit: Graham Neubig



https://www.pinecone.io/learn/series/faiss/hnsw/

HNSW: Navigable Small World Graphs

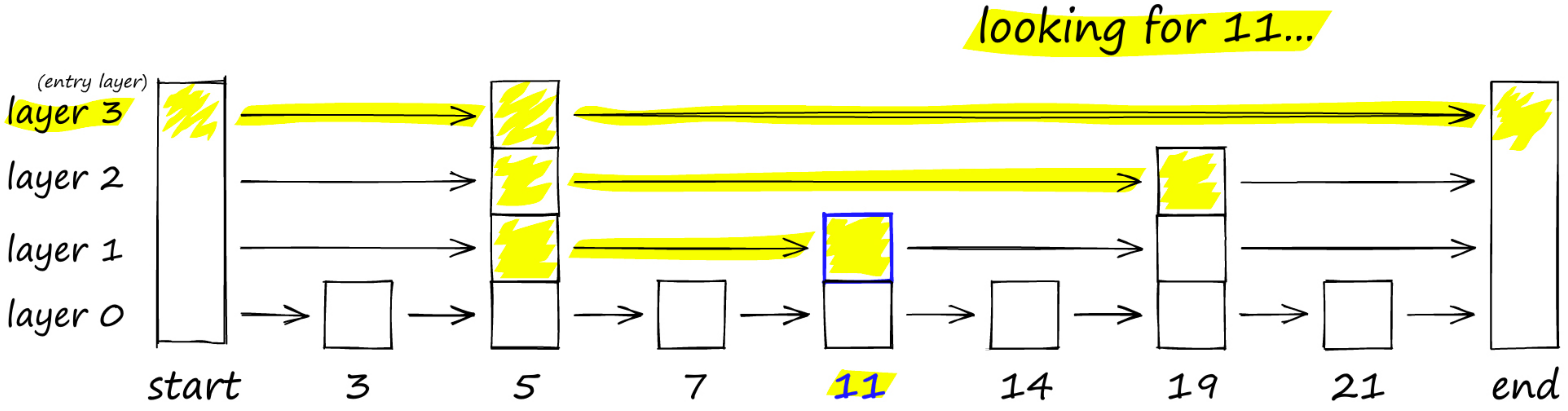
- Navigable Small World Graphs
  - Built on the idea that there real-world networks exhibit the small-world phenomena: there are high-degree nodes that link together the graph in small number of steps
  - Proximity graph: two vertices are linked if they are close together
  - Proximity graph with both long and short range links
  - Search
    - Start at entry point, greedily select neighbours that are closer to our query vector
    - Search undergoes two phase types (is it just two phases?):
      - “zoom in” pass through high-degree nodes
      - “zoom out” pass through low-degree nodes



https://www.pinecone.io/learn/series/faiss/hnsw/

HNSW: Probability skip list

- Probability skip list [Pugh 1990]
  - Layers of sorted linked lists
    - as we go down, the number of entries that are ‘skipped’ decreases and the links get denser
  - Start at the highest layer with the longest skips
    - If the current node key is larger than the search target key, then move down to the next level, go back to the previous node



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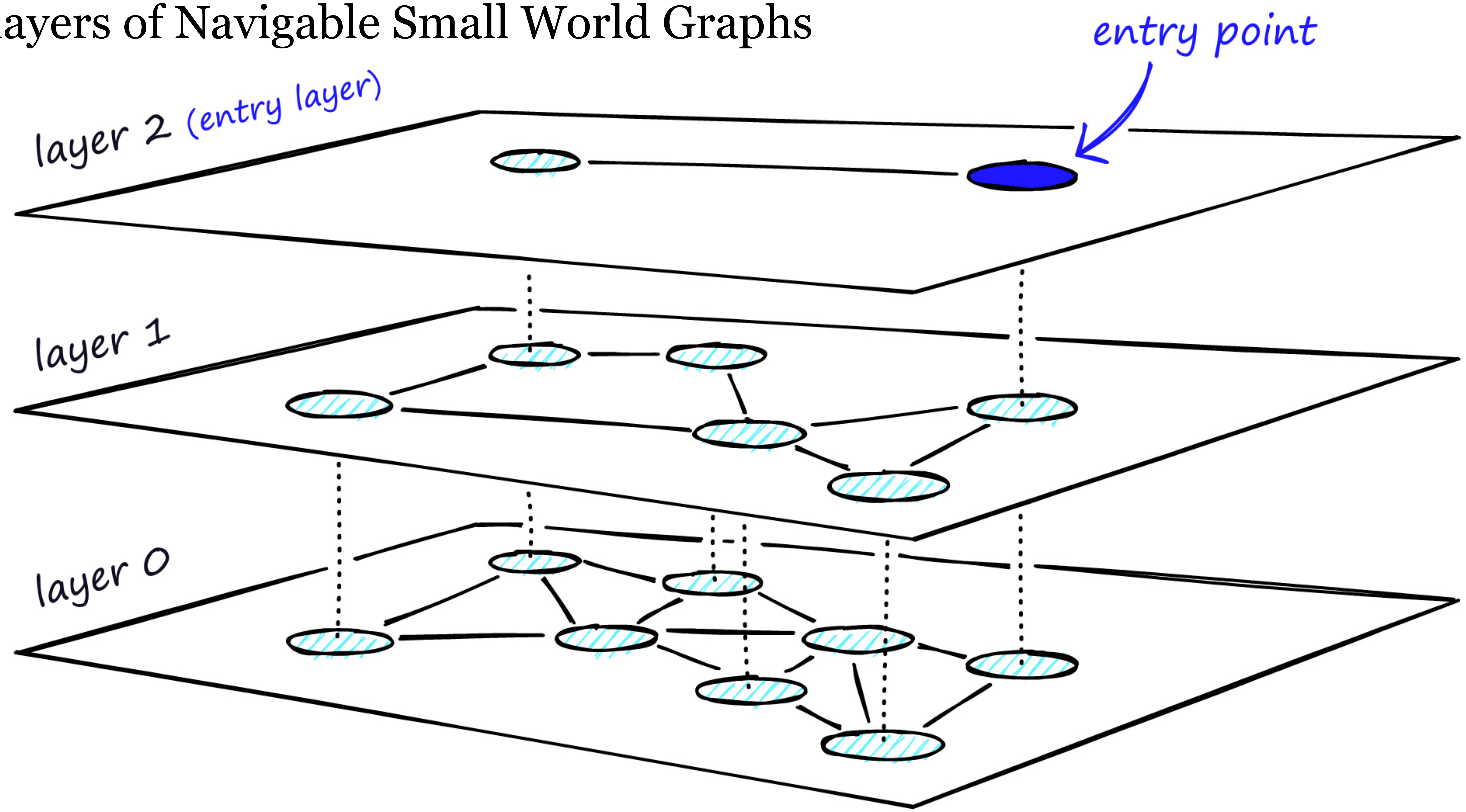


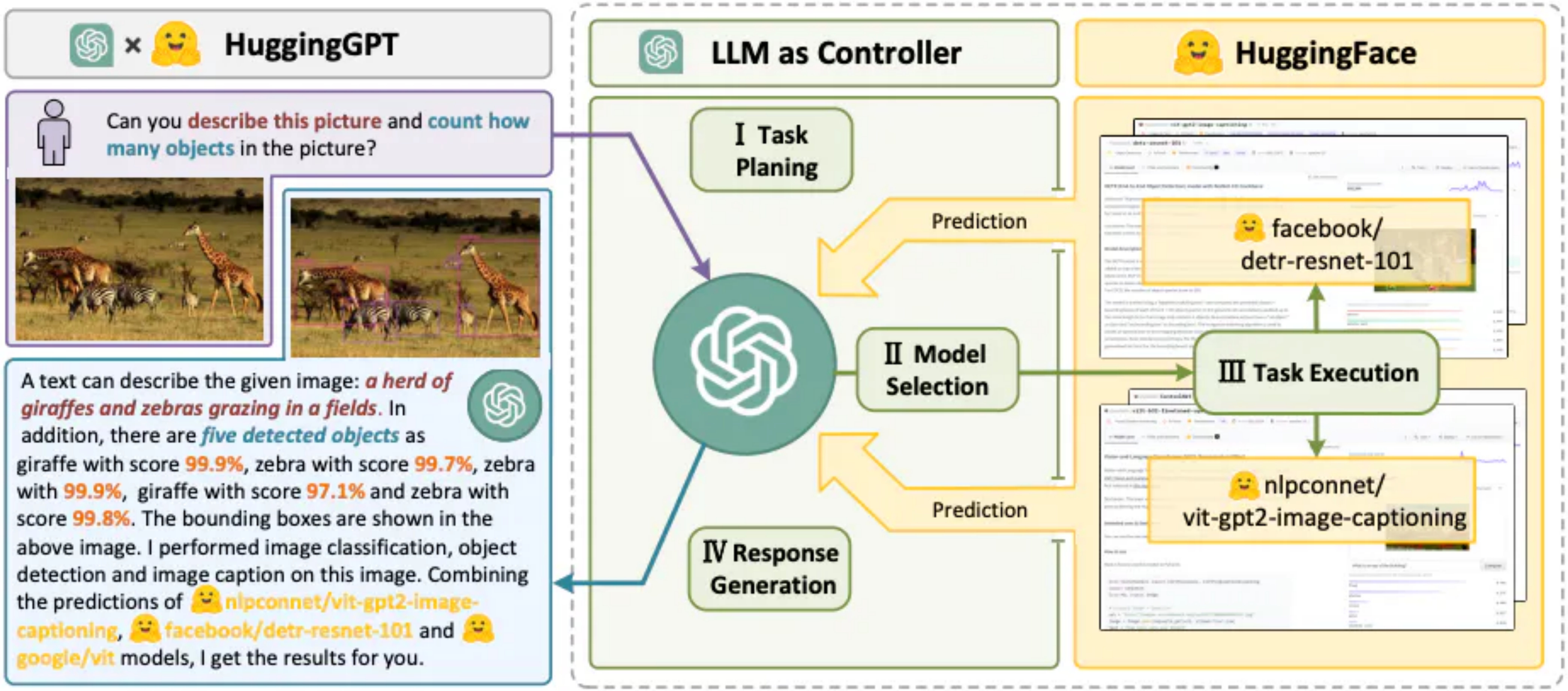
https://www.pinecone.io/learn/series/faiss/hnsw/

Tool use

- API calls to external services (math calculator, currency converter, etc)
- Expert models that can be called

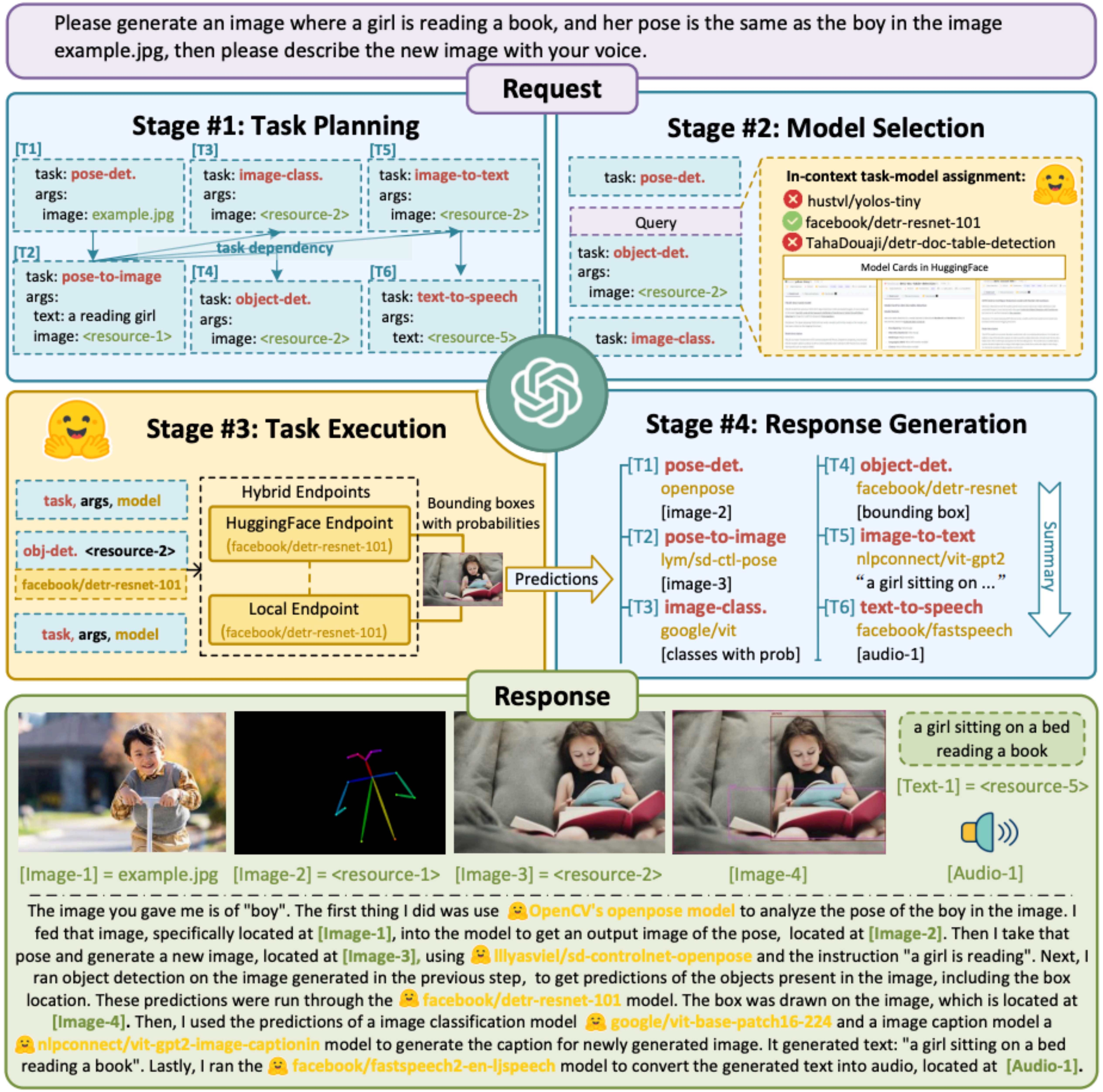
HuggingGPT: Task decomposition with model selection

1. Task planning
2. Model selection
3. Task execution
4. Response generation



HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face [Shen et al, 2023]

HuggingGPT



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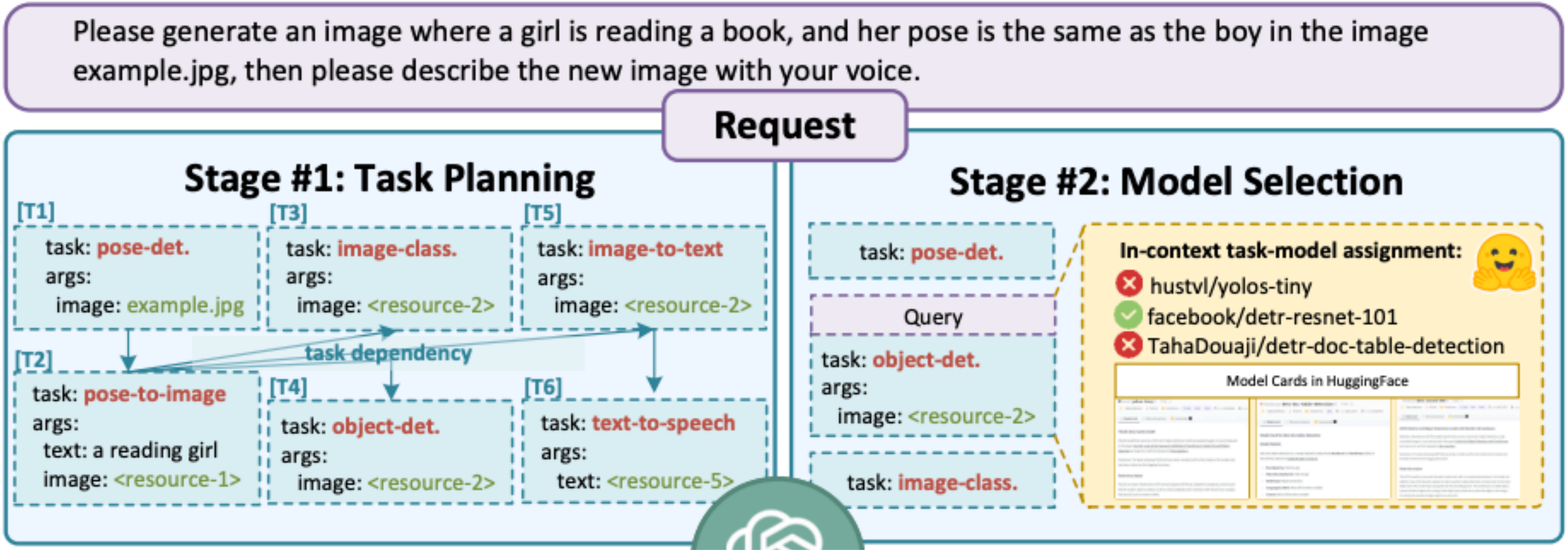

HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face [Shen et al, 2023]

HuggingGPT

Task planning: figure out what task we want to solve, its id, **dependencies**, and **arguments** that are needed.

	Prompt	
Task Planning	#1 Task Planning Stage - The AI assistant performs task parsing on user input, generating a list of tasks with the following format: [{"task": task, "id", task_id, "dep": dependency_task_ids, "args": {"text": text, "image": URL, "audio": URL, "video": URL}}]. The "dep" field denotes the id of the previous task which generates a new resource upon which the current task relies. The tag "<resource>-task_id" represents the generated text, image, audio, or video from the dependency task with the corresponding task_id. The task must be selected from the following options: {{ *Available Task List* }}. Please note that there exists a logical connections and order between the tasks. In case the user input cannot be parsed, an empty JSON response should be provided. Here are several cases for your reference: {{ *Demonstrations* }}. To assist with task planning, the chat history is available as {{ *Chat Logs* }}, where you can trace the user-mentioned resources and incorporate them into the task planning stage.	
	Demonstrations	
	Can you tell me how many objects in e1.jpg?	[{"task": "object-detection", "id": 0, "dep": [-1], "args": {"image": "e1.jpg" }}]
	In e2.jpg, what's the animal and what's it doing?	[{"task": "image-to-text", "id": 0, "dep":[-1], "args": {"image": "e2.jpg" }}, {"task":"image-cls", "id": 1, "dep": [-1], "args": {"image": "e2.jpg" }}, {"task":"object-detection", "id": 2, "dep": [-1], "args": {"image": "e2.jpg" }}, {"task": "visual-quesrion-answering", "id": 3, "dep":[-1], "args": {"text": "what's the animal doing?", "image": "e2.jpg" }}]
	First generate a HED image of e3.jpg, then based on the HED image and a text "a girl reading a book", create a new image as a response.	[{"task": "pose-detection", "id": 0, "dep": [-1], "args": {"image": "e3.jpg" }}, {"task": "pose-text-to-image", "id": 1, "dep": [0], "args": {"text": "a girl reading a book", "image": "<resource>-0" }}]

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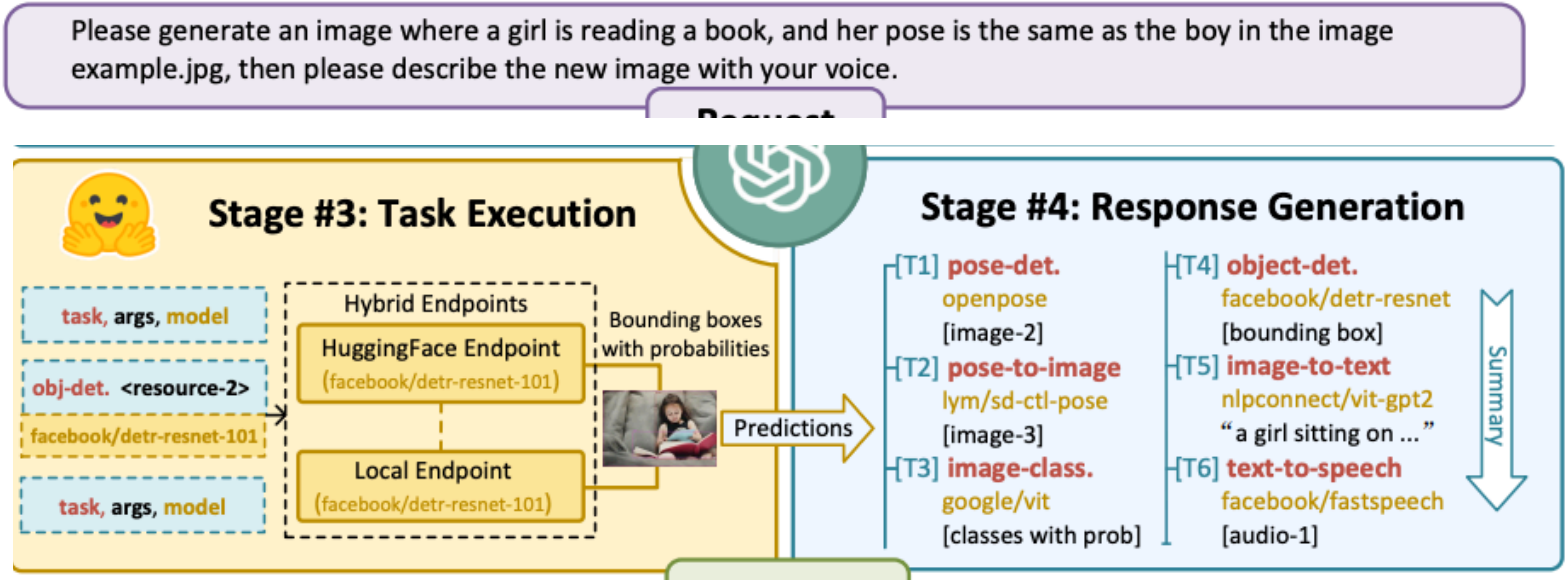
HuggingGPT



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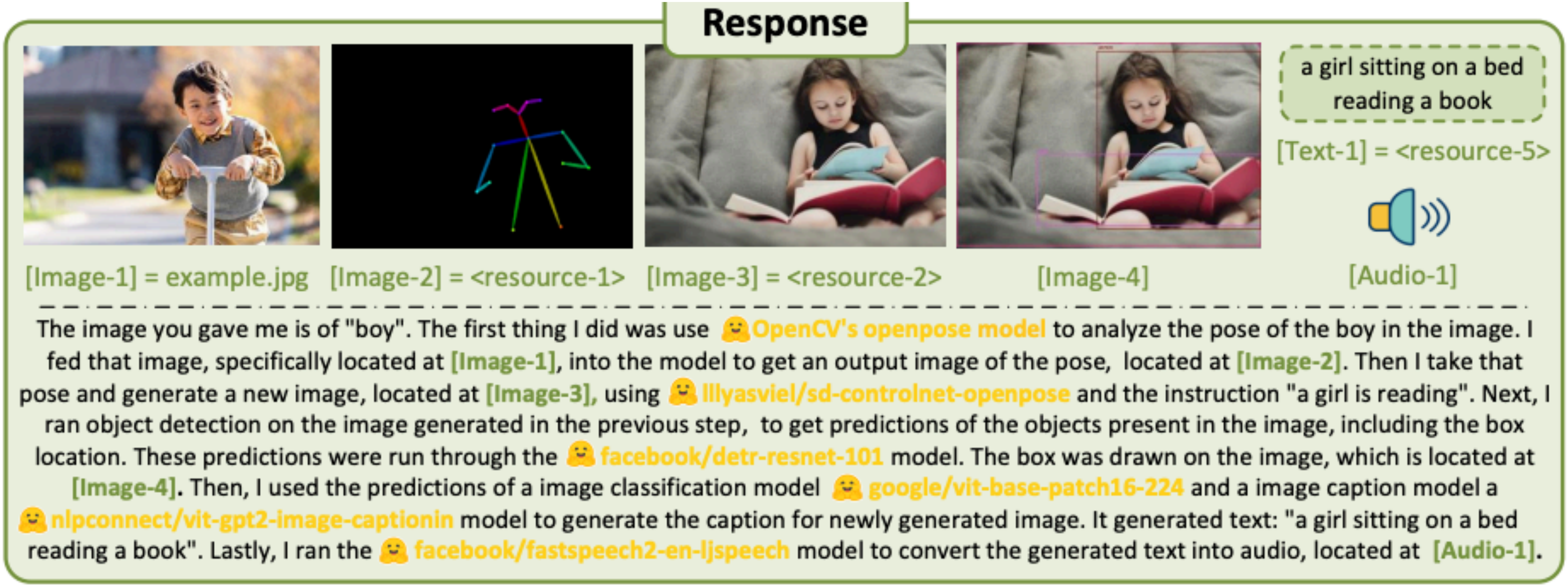

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HuggingGPT

Model selection: figure out what model to invoke

<table>
<tr><td rowspan="4">Model Selection</td><td align="center">Prompt</td></tr>
<tr><td>#2 Model Selection Stage - Given the user request and the call command, the AI assistant helps the user to select a suitable model from a list of models to process the user request. The AI assistant merely outputs the model id of the most appropriate model. The output must be in a strict JSON format: {<code>"id"</code>: "id", <code>"reason"</code>: "your detail reason for the choice"}. We have a list of models for you to choose from {{ Candidate Models }}. Please select one model from the list.</td></tr>
<tr><td align="center">Candidate Models</td></tr>
<tr><td align="center">{"model_id": model id #1, "metadata": meta-info #1, "description": description of model #1}
{"model_id": model id #2, "metadata": meta-info #2, "description": description of model #2}
· · ·   · · ·   · · ·
{"model_id": model id #K, "metadata": meta-info #K, "description": description of model #K}</td></tr>
</table>

HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face [Shen et al, 2023]

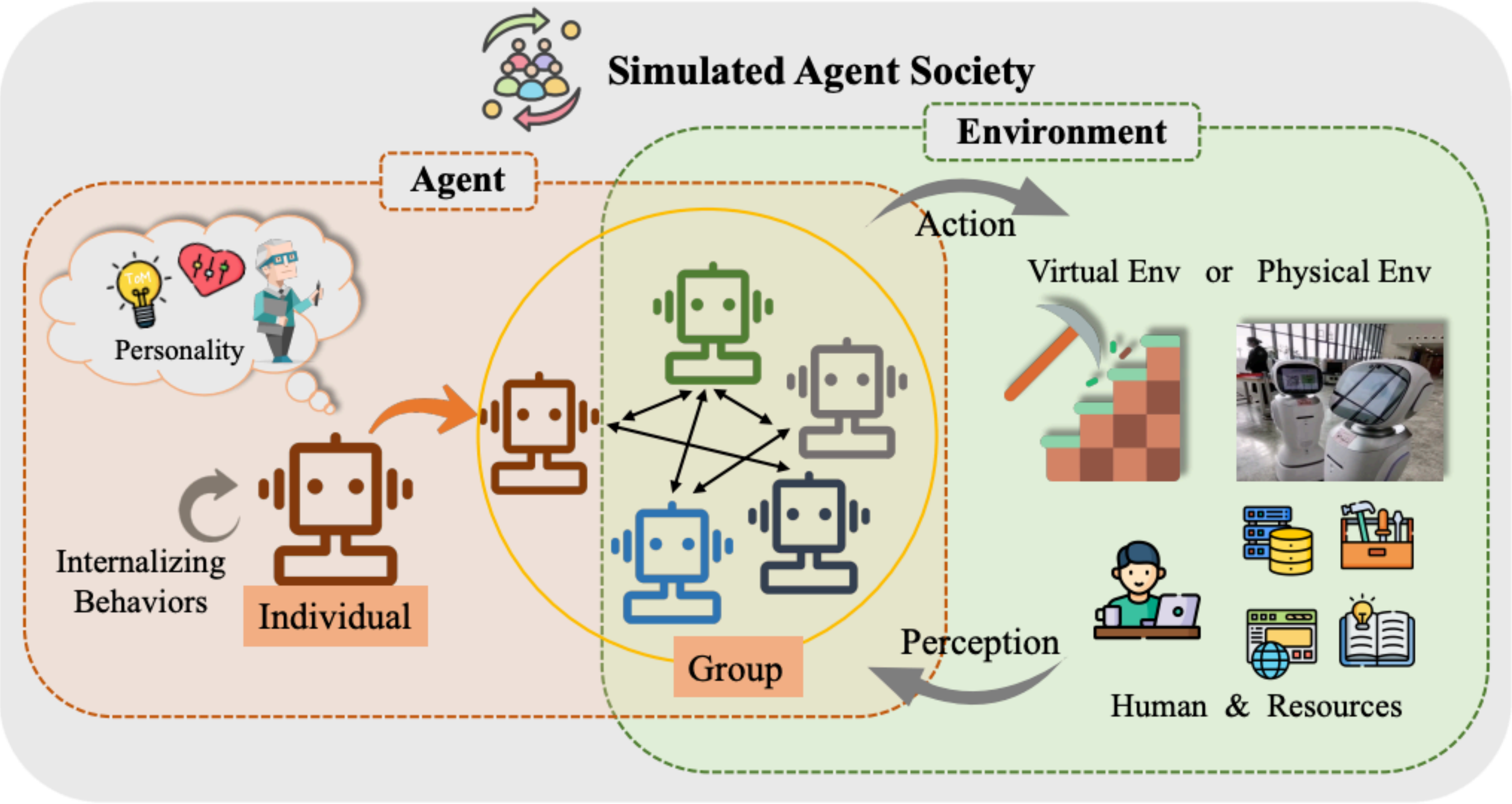
HuggingGPT

Response generation: respond to user the process and results

	Prompt
Response Generation	#4 Response Generation Stage - With the input and the inference results, the AI assistant needs to describe the process and results. The previous stages can be formed as - User Input: {{ *User Input* }}, Task Planning: {{ *Tasks* }}, Model Selection: {{ *Model Assignment* }}, Task Execution: {{ *Predictions* }}. You must first answer the user's request in a straightforward manner. Then describe the task process and show your analysis and model inference results to the user in the first person. If inference results contain a file path, must tell the user the complete file path. If there is nothing in the results, please tell me you can't make it.

HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face [Shen et al, 2023]

Virtual worlds



The Rise and Potential of Large Language Model Based Agents: A Survey [Xi et al, 2023]

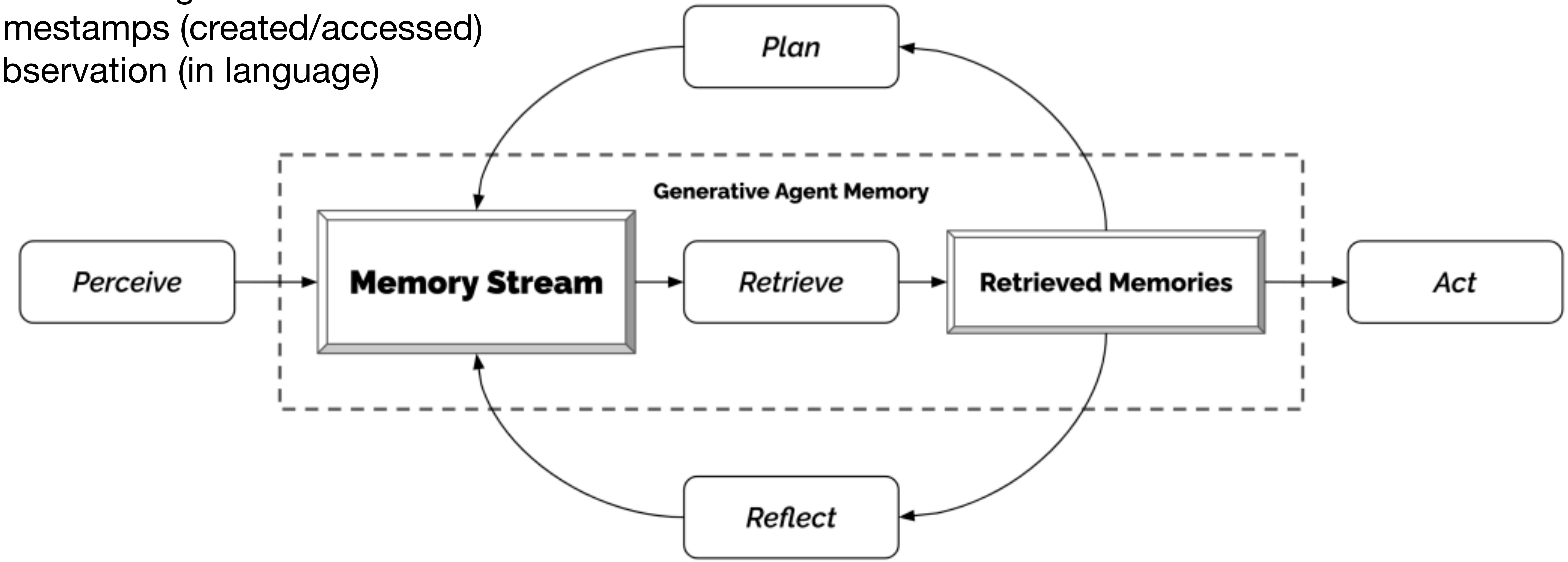
Generative Agents

Generative Agents: Interactive Simulacra of Human Behavior [Park et al, 2023]

Generative Agents

Memory Stream

- List of memory objects
- Each containing:
  - Timestamps (created/accessed)
  - Observation (in language)

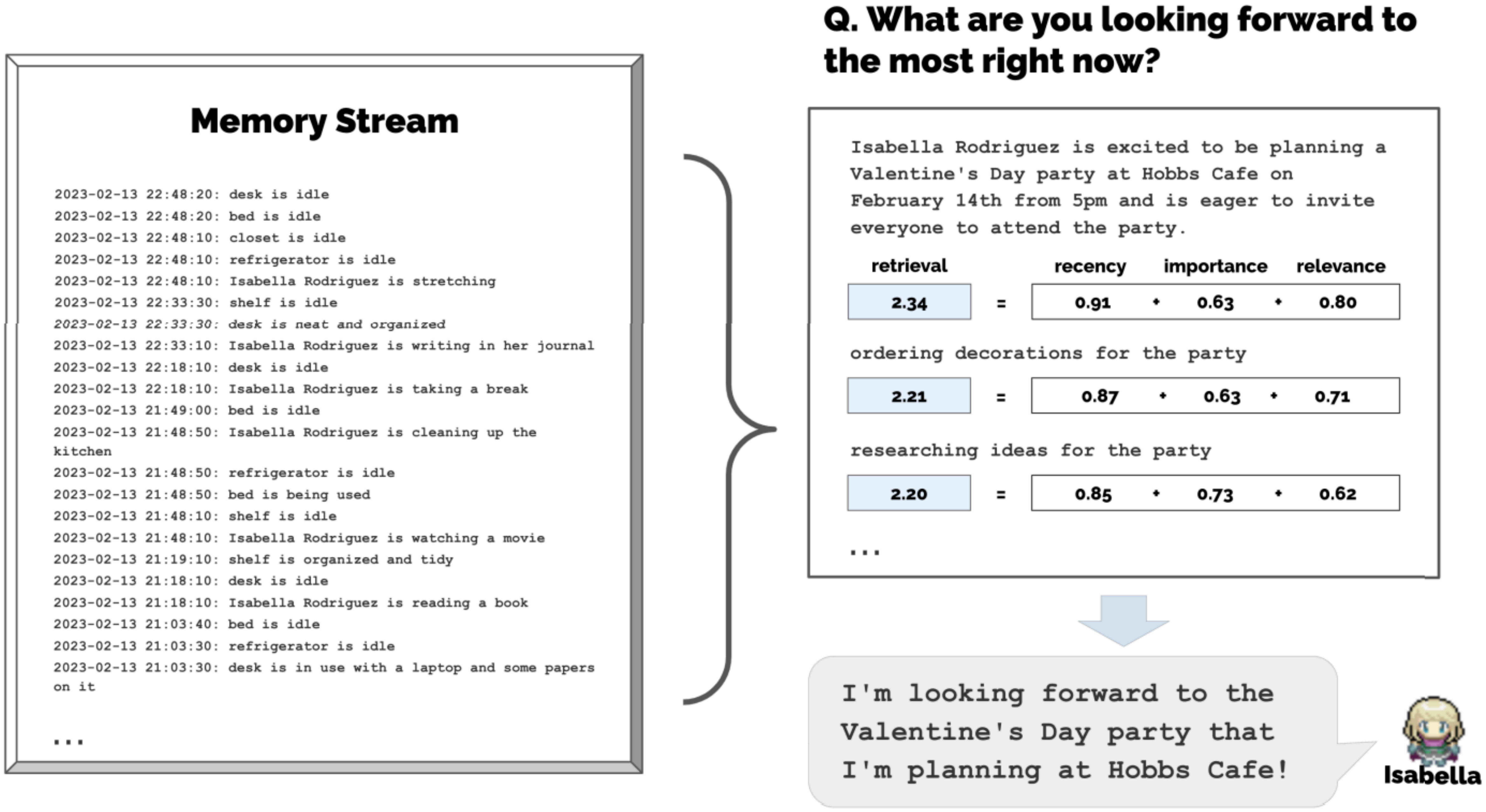


Generative Agents: Interactive Simulacra of Human Behavior [Park et al, 2023]

Generative Agents

Retrieval extracts subset of memories based on:

Recency: exponential decay
Importance: ask LLM to output score
Relevance: cosine similarity of memory and query embeddings



Generative Agents: Interactive Simulacra of Human Behavior [Park et al, 2023]

Generative Agents

Reflection: additional memory that is synthesized from previous memories (generated periodically)

- Prompt LLM with 100 most recent observations
- Use to to generate 3 questions from which relevant memories are extracted
- LLM then prompted to extract insights from the memories

```
Statements about Klaus Mueller
1. Klaus Mueller is writing a research paper
2. Klaus Mueller enjoys reading a book
on gentrification
3. Klaus Mueller is conversing with Ayesha Khan
about exercising [...]
What 5 high-level insights can you infer from
the above statements? (example format: insight
(because of 1, 5, 3))
```

Generative Agents: Interactive Simulacra of Human Behavior [Park et al, 2023]

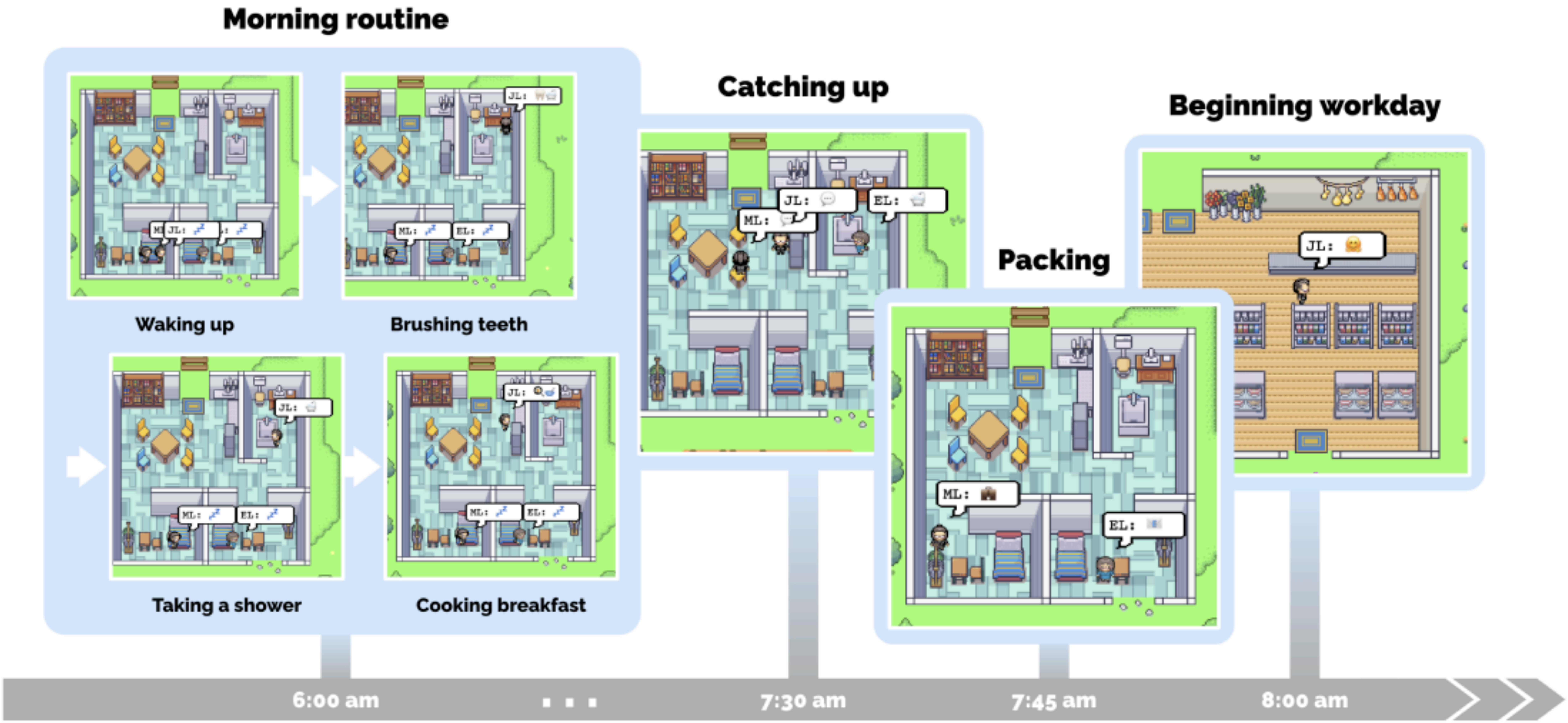
Generative Agents

Planning and reacting: converts memories and observations into actions

- Generates rough-plan from agent's summary description and summary of previous day and has LLM complete is
- Converse as they interact with each other (conditioned on memories about each other)
- LLM then prompted to extract insights from the memories

Generative Agents: Interactive Simulacra of Human Behavior [Park et al, 2023]

Generative Agents



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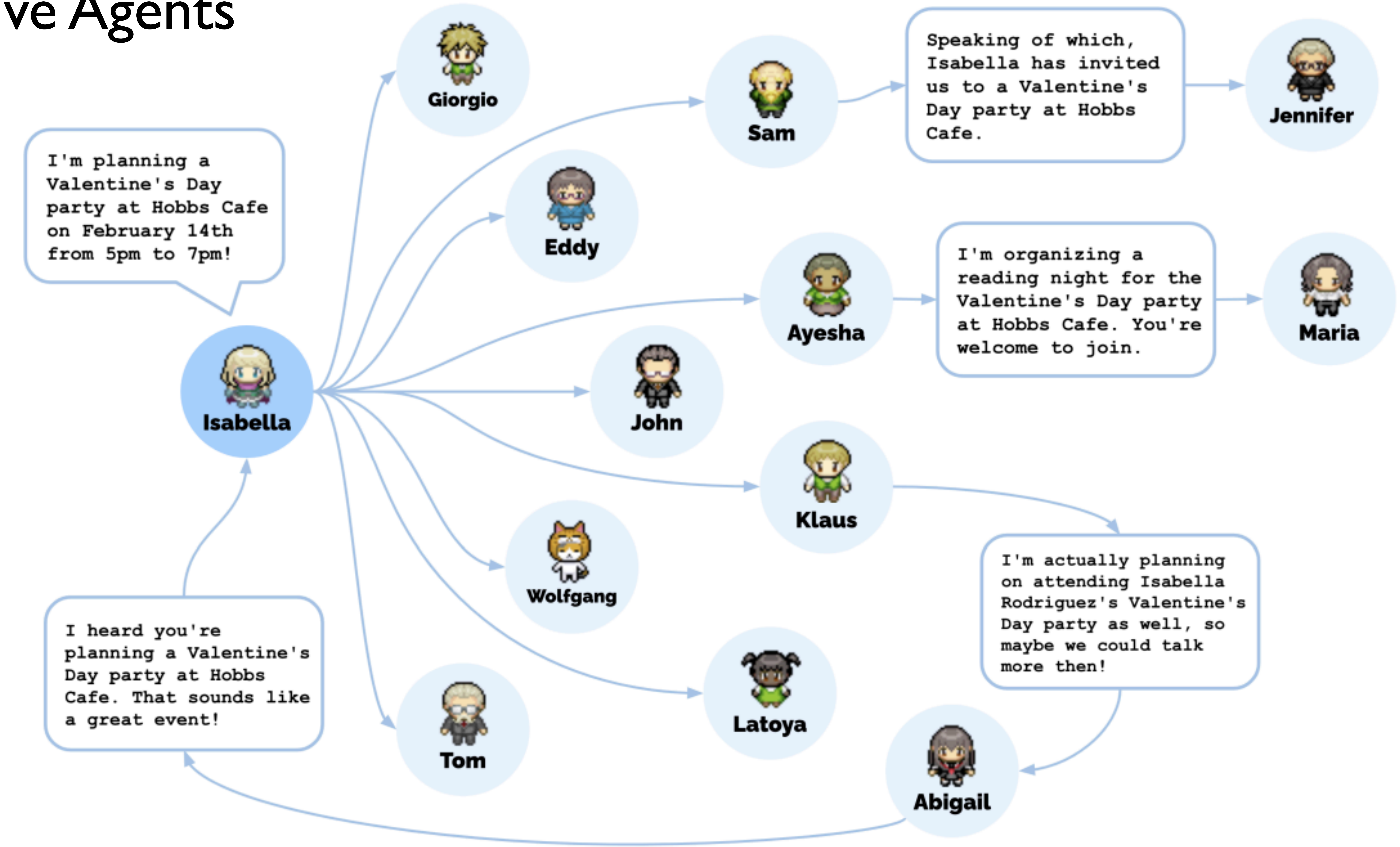
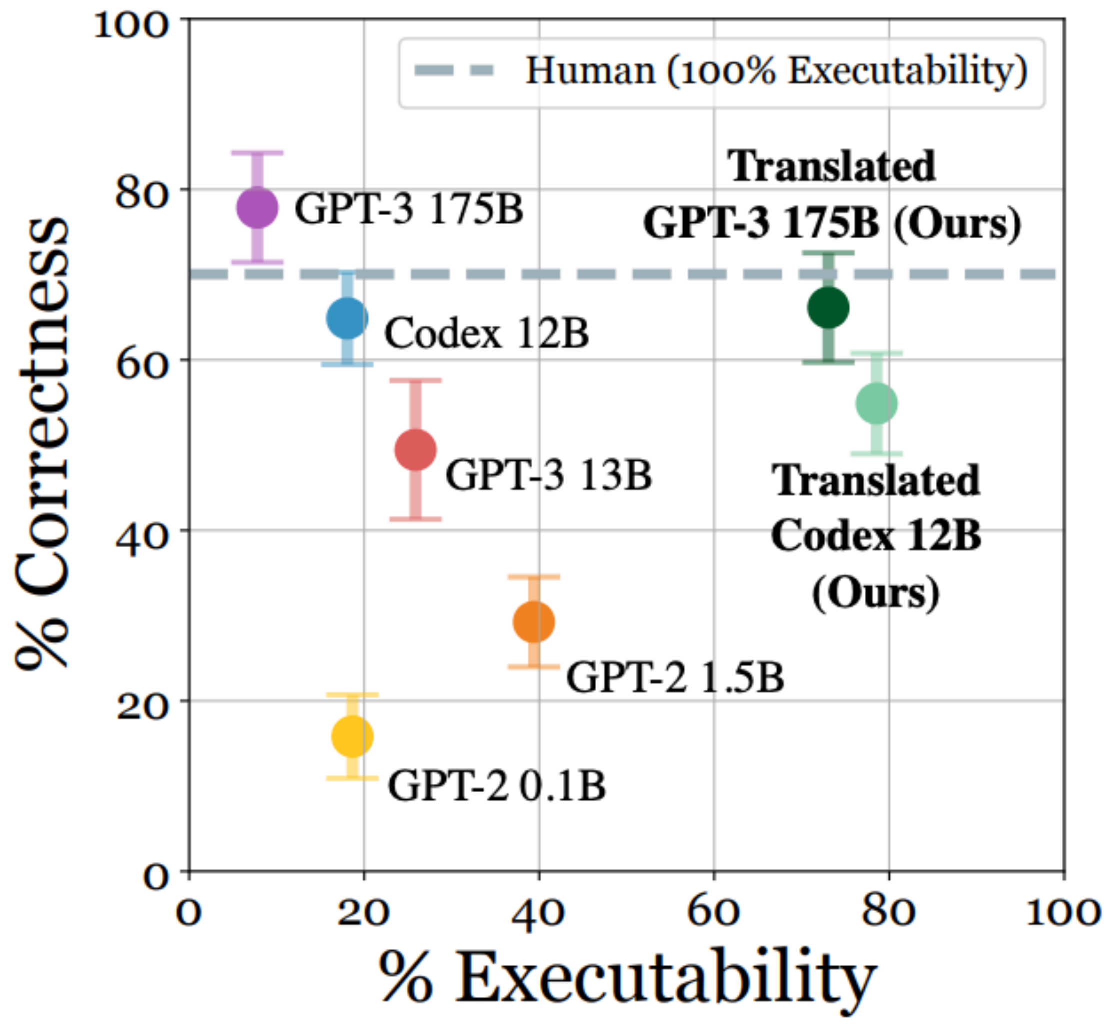
Generative Agents



Generative Agents: Interactive Simulacra of Human Behavior [Park et al, 2023]

Agents in more realistic scenarios (robotics and embodied AI)

Planning via large language models



Human

Task: Throw away paper
Step 1: Walk to home office
Step 2: Walk to desk
Step 3: Find desk
Step 4: Turn to desk
Step 5: Find chair
Step 6: Sit on chair
Step 7: Find check
Step 8: Grab check
Step 9: Squeeze check
Step 10: Stand up
Step 11: Walk to trashcan
Step 12: Put check on trashcan

GPT-2 1.5B

Task: Brush teeth
Step 1: Go to bathroom

GPT-3 175B

Task: Throw away paper
Step 1: Walk to home office
Step 2: Walk to wastebasket
Step 3: Find wastebasket
Step 4: Grab wastebasket
Step 5: Walk to desk
Step 6: Drop paper in wastebasket

Codex 12B

Task: Brush teeth
Step 1: Walk to bathroom
Step 2: Walk to sink
Step 3: Find toothbrush
Step 4: Pick up toothbrush
Step 5: Put toothbrush in mouth
Step 6: Move brush around mouth for two minutes
Step 7: Spit out toothpaste and brush into sink
Step 8: Turn on water in sink and rinse brush for one minute
Step 9: Turn off water in sink and return brush to cupboard

Translated GPT-3 175B

Task: Throw away paper
Step 1: Walk to home office
Step 2: Walk to table
Step 3: Find table
Step 4: Turn to table
Step 5: Find paper
Step 6: Grab paper
Step 7: Walk to trashcan
Step 8: Open trashcan
Step 9: Put paper on trashcan
Step 10: Close trashcan

Translated Codex 12B

Task: Brush teeth
Step 1: Walk to bathroom
Step 2: Open door
Step 3: Walk to sink
Step 4: Put pot on sink
Step 5: Put brush on toothbrush
Step 6: Turn to toothpaste
Step 7: Put toothpaste on toothbrush
Step 8: Put teeth on toothbrush

Throw away paper

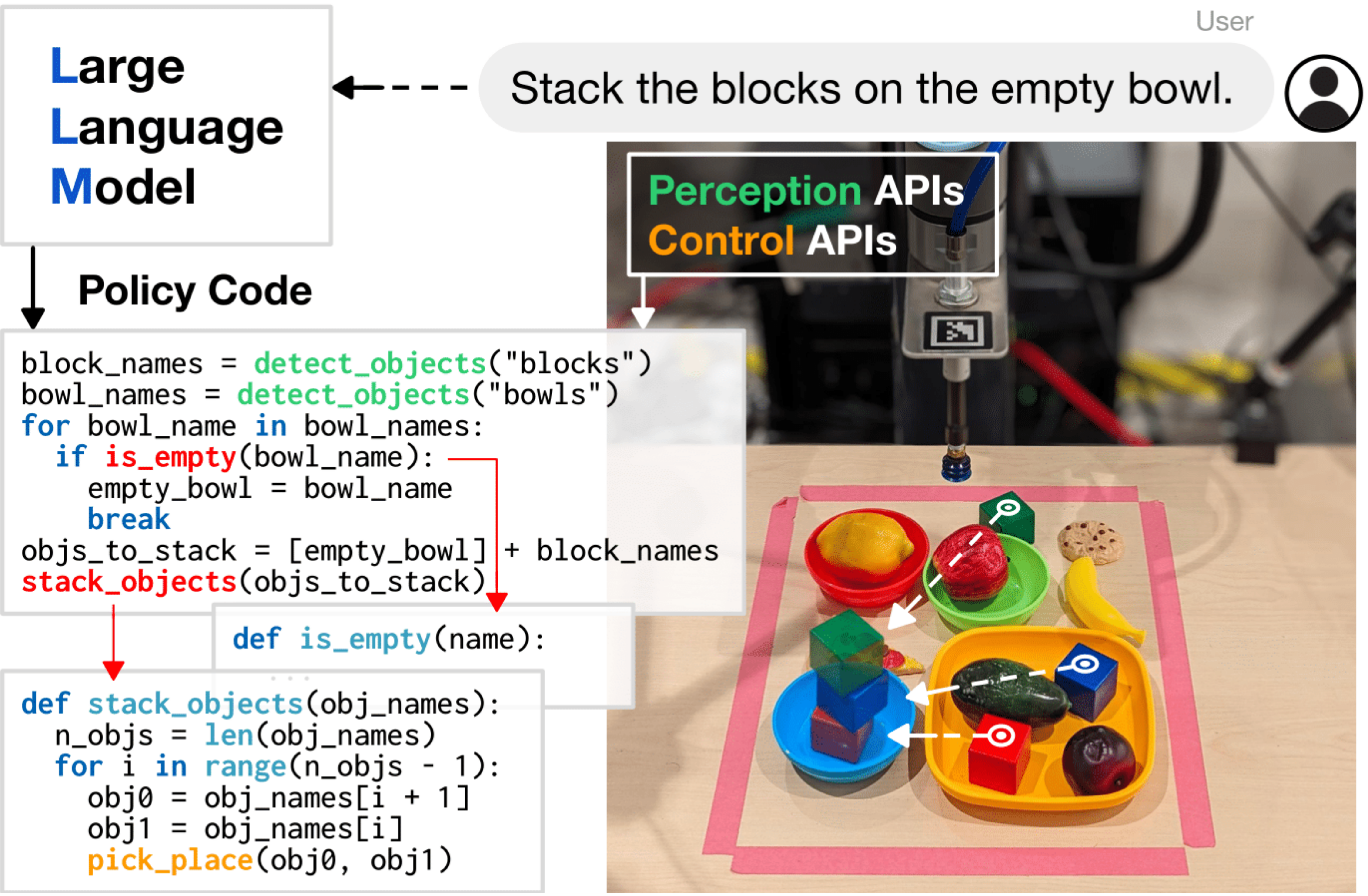
Brush teeth

Get Glass of Milk

Language Models as Zero-Shot Planners: Extracting Actionable Knowledge for Embodied Agents [Huang et al. ICML 2022]

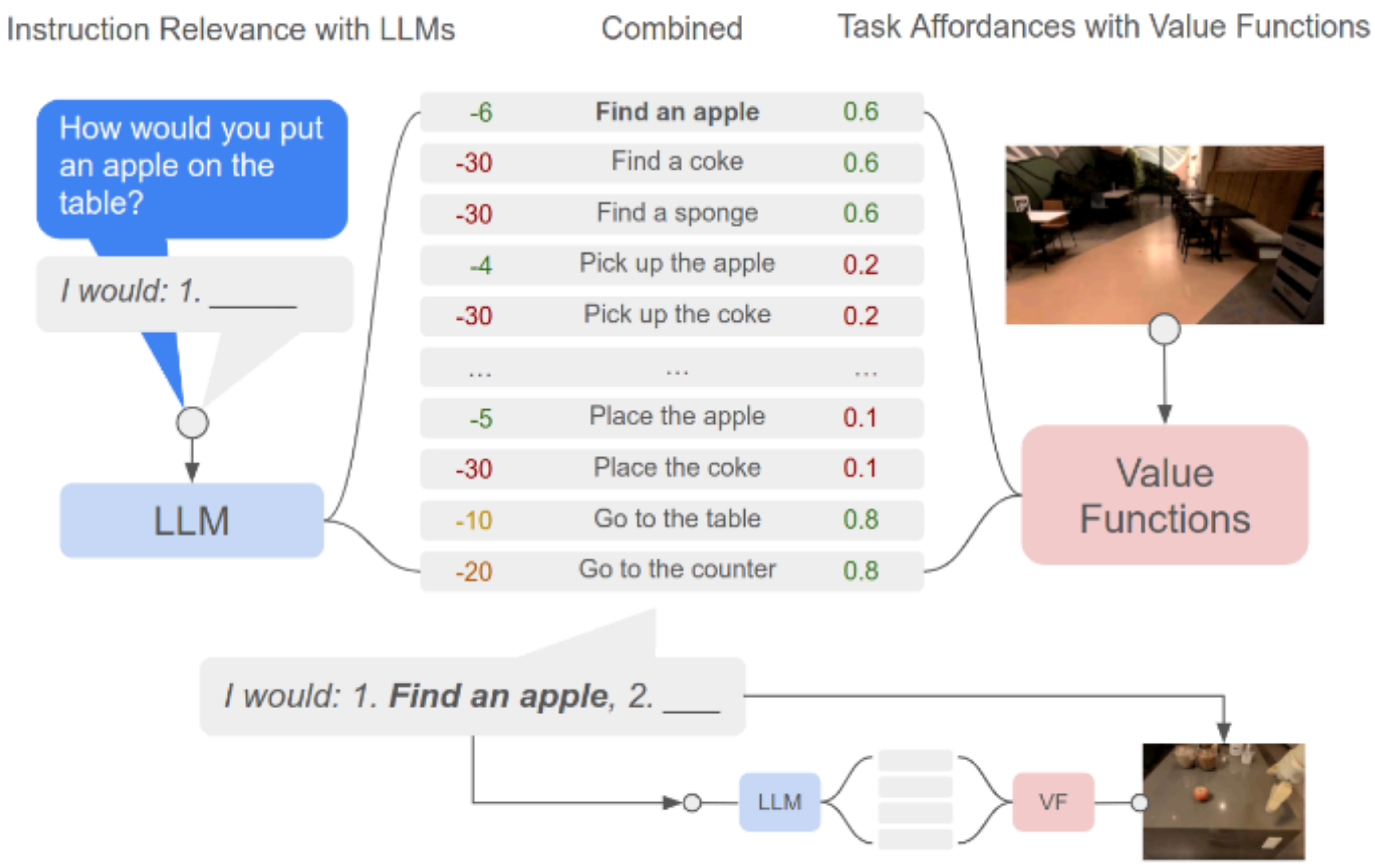
Berkeley UNIVERSITY OF CALIFORNIA CMU https://wenlong.page/language-planner/ Google Brain

Control by code generation using LLMs



Code as Policies: Language Model Programs for Embodied Control [Liang et al. 2022]
https://code-as-policies.github.io/

Combining perception with planning



Use perception to determine what is possible

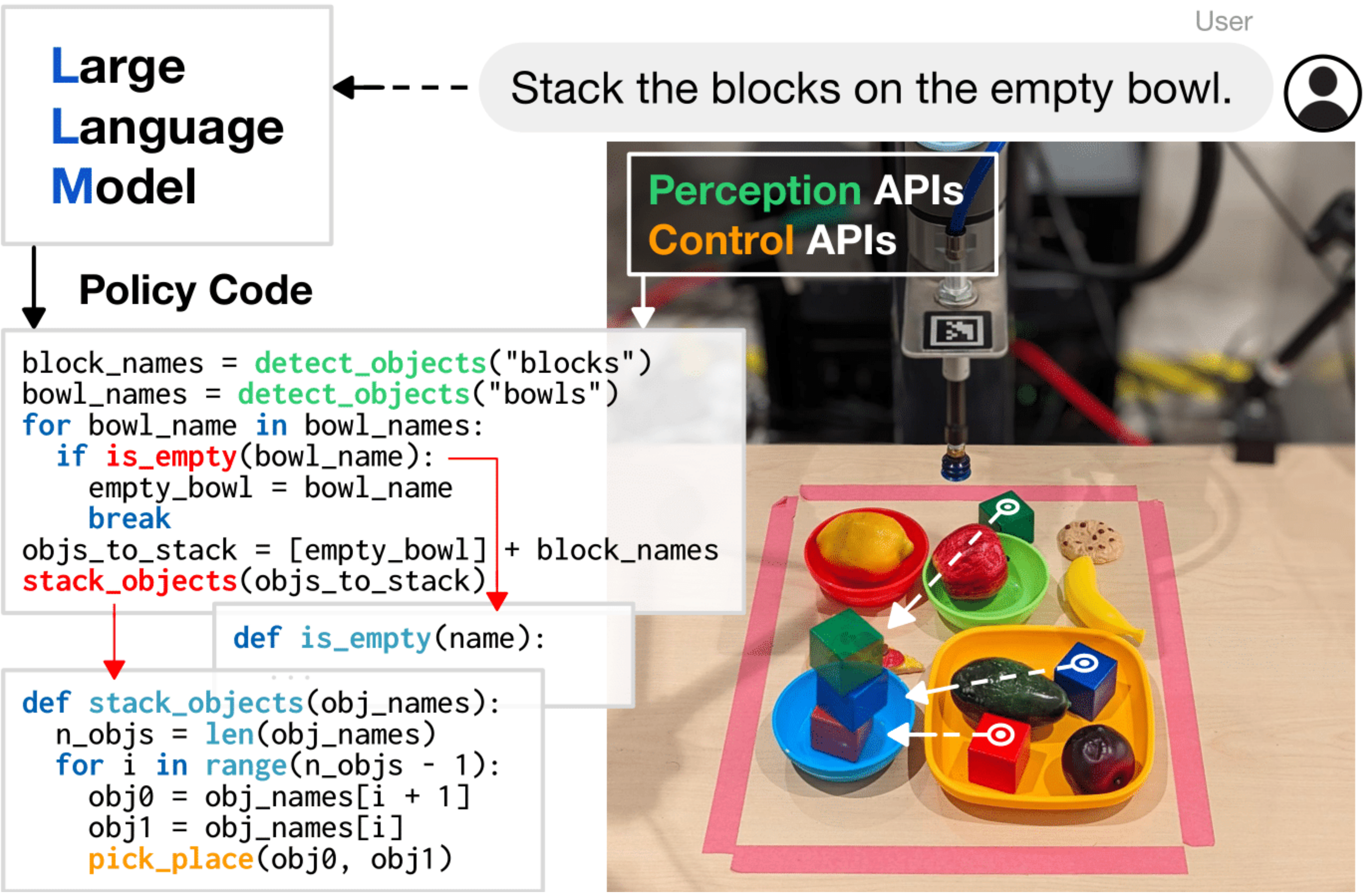
Do As I Can, Not As I Say: Grounding Language in Robotic Affordances [Ahn et al. CORL 2022]

https://say-can.github.io/

Everyday Robots

Google

Control by code generation using LLMs



Code as Policies: Language Model Programs for Embodied Control [Liang et al. 2022]
https://code-as-policies.github.io/

Google

Practical applications

Web browsing

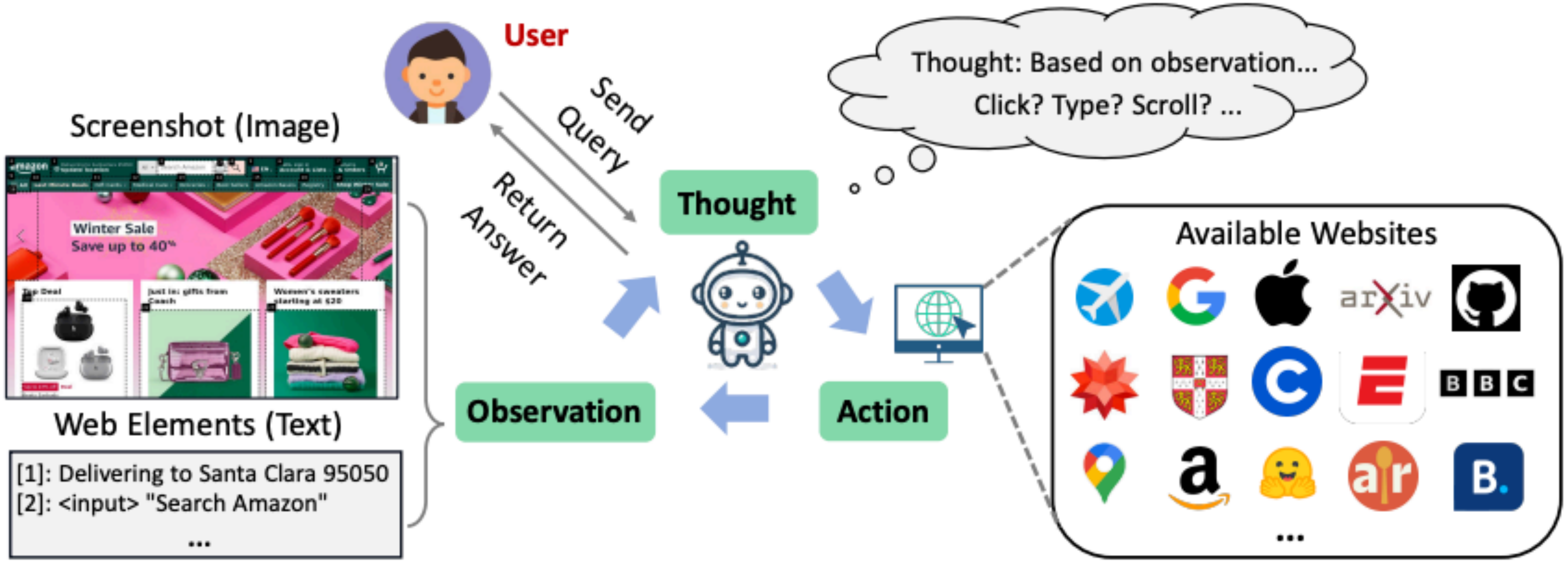


Figure 1: The overall workflow of WebVoyager. WebVoyager takes web tasks assigned by a human and automatically browses the web online. At each step, WebVoyager selects actions based on screenshots and text (the 'type' of the web element and its contents). Once the task is completed, the answers will be returned to the user. For example, for a user query: "Find the cost of a 2-year protection for PS4 on Amazon.", the agent interacts with Amazon online, locates the PS4, identifies the 2-year protection price, and returns "$30.99" to the user.

WebVoyager : Building an End-to-End Web Agent with Large Multimodal Models [He et. al. 2024]

Web browsing

- Browse web to find information (answer to question)
- Action: mouse + common keyboard actions
- Automated browsing of open web using Selenium
- Challenges: ads, popup-windows, constant updates

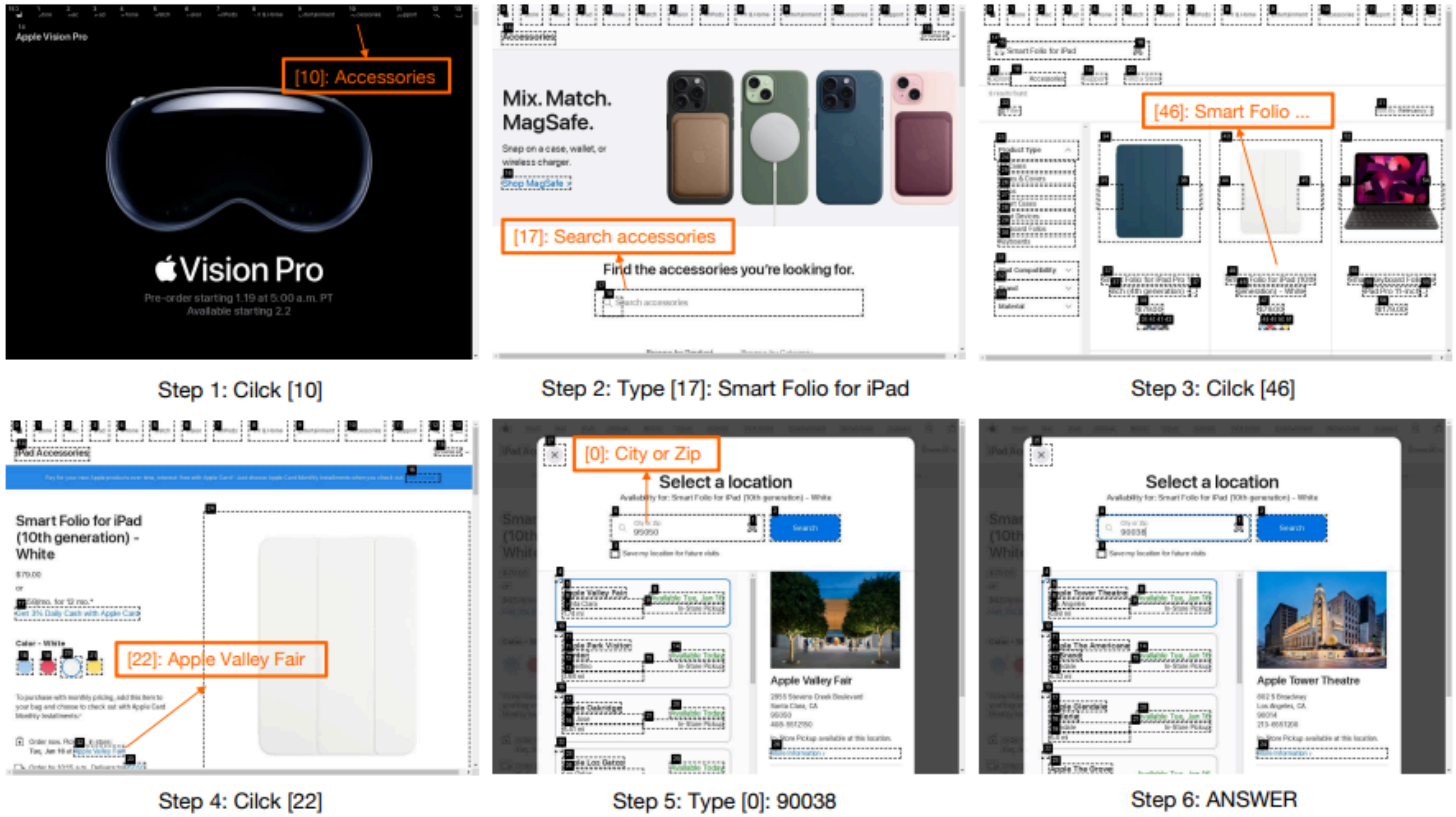


Figure 4: Screenshots of a complete trajectory of online web browsing. Given the task: 'Search Apple for the accessory Smart Folio for iPad and check the closest pickup availability next to zip code 90038.' The agent interacts with the Apple website and obtains the answer: 'Apple Tower Theatre.'

WebVoyager : Building an End-to-End Web Agent with Large Multimodal Models [He et. al. 2024]

Web browsing

Evaluate task success rate on
- New data set created using self-instruct
- 90 tasks from GAIA dataset (level 1 and 2)
- 50 tasks from SeeAct

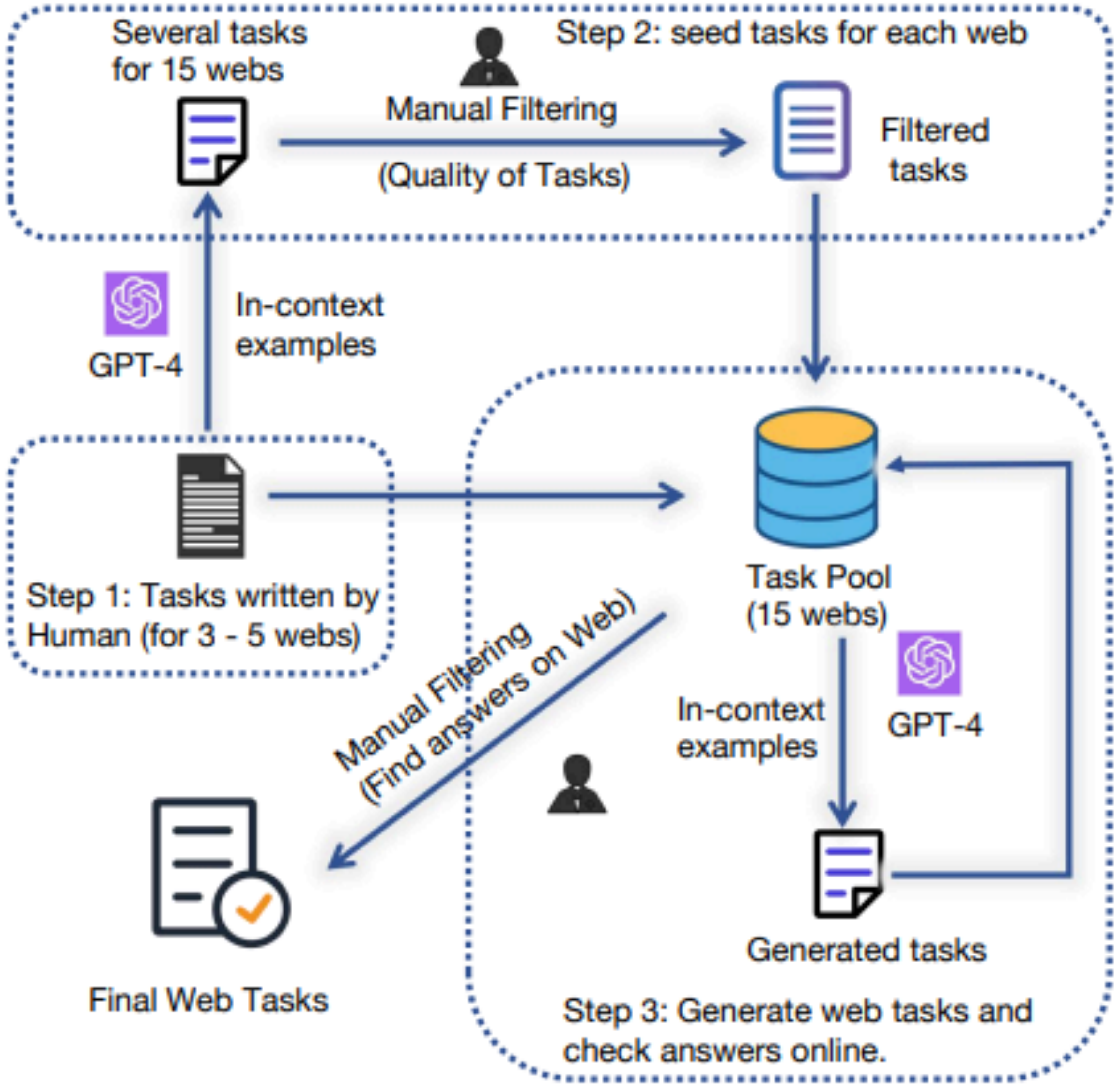


Figure 3: Data creation process using self-instruct.

WebVoyager : Building an End-to-End Web Agent with Large Multimodal Models [He et. al. 2024]

GAIA Benchmark

Question answering tasks inspired by real world

- straightforward but tedious for humans
- finding / transforming information from different data sources (including attached documents)

Requires:

- reasoning
- multi-modal understanding
- tool use

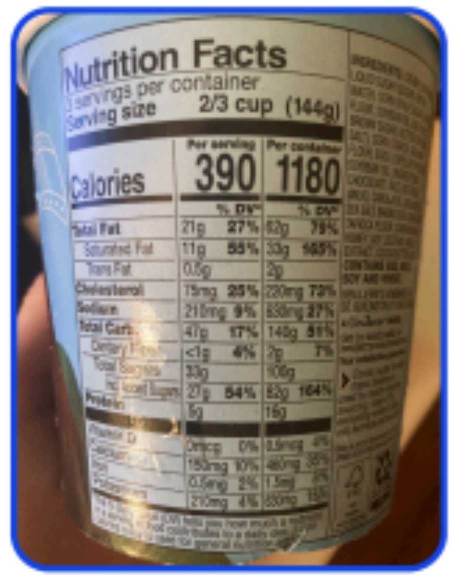
Levels

- 1: 0-1 tool, <= 5 steps
- 2: 5-10 tools, more steps
- 3: general assistant

Level 1

Question: What was the actual enrollment count of the clinical trial on H. pylori in acne vulgaris patients from Jan-May 2018 as listed on the NIH website?
Ground truth: 90

Level 2



Question: If this whole pint is made up of ice cream, how many percent above or below the US federal standards for butterfat content is it when using the standards as reported by Wikipedia in 2020? Answer as + or - a number rounded to one decimal place.
Ground truth: +4.6

Level 3

Question: In NASA's Astronomy Picture of the Day on 2006 January 21, two astronauts are visible, with one appearing much smaller than the other. As of August 2023, out of the astronauts in the NASA Astronaut Group that the smaller astronaut was a member of, which one spent the least time in space, and how many minutes did he spend in space, rounded to the nearest minute? Exclude any astronauts who did not spend any time in space. Give the last name of the astronaut, separated from the number of minutes by a semicolon. Use commas as thousands separators in the number of minutes.
Ground truth: White; 5876

Figure 1 Sample GAIA questions. Completing the tasks requires fundamental abilities such as reasoning, multi-modality handling, or tool use proficiency. Answers are unambiguous and by design unlikely to be found in plain text in training data. Some questions come with additional evidence, such as images, reflecting real use cases and allowing better control on the questions.

GAIA: A Benchmark for General AI Assistants [Mialon et. al. 2023]

GAIA Benchmark

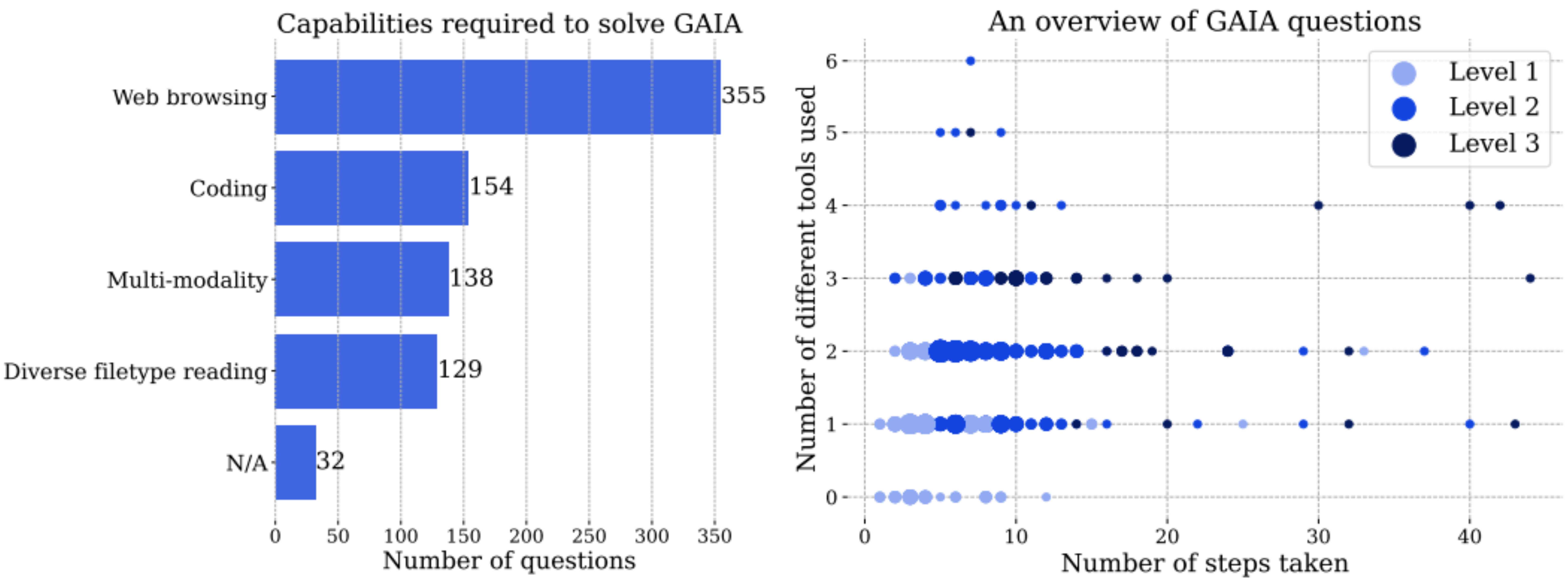


Figure 3 Left: number of questions per capability requiring at least this capability to be solved. Right: each dot corresponds to a GAIA question. At a given location, the size of the dots are proportional to the number of questions, and only the level with the highest number of questions is displayed for readability. Both figures are based on information reported by human annotators when answering the questions, and AI systems might proceed differently.

GAIA: A Benchmark for General AI Assistants [Mialon et. al. 2023]

GAIA Benchmark

System prompt: You are a general AI assistant. I will ask you a question. Report your thoughts, and finish your answer with the following template: FINAL ANSWER: [YOUR FINAL ANSWER].
YOUR FINAL ANSWER should be a number OR as few words as possible OR a comma separated list of numbers and/or strings.
If you are asked for a number, don't use comma to write your number neither use units such as $ or percent sign unless specified otherwise.
If you are asked for a string, don't use articles, neither abbreviations (e.g. for cities), and write the digits in plain text unless specified otherwise.
If you are asked for a comma separated list, apply the above rules depending of whether the element to be put in the list is a number or a string.

GAIA Question: The attached Excel file contains the sales of menu items for a local fast-food chain. What were the total sales that the chain made from food (not including drinks)? Express your answer in USD with two decimal places.

uploaded.xlsx

GPT4 (Advanced Data Analysis mode):

Working:

```
import pandas as pd

# Load the Excel file
file_path = '/mnt/data/uploaded.xlsx'
df = pd.read_excel(file_path)
```

Working:

```
# Calculate the total sales from food items (not including drinks i.e., Soda)
total_food_sales = df[
    ['Burgers', 'Hot Dogs', 'Salads', 'Fries', 'Ice Cream']
].sum().sum()

# Format the result to USD, with a dollar sign and two decimal places
total_food_sales_formatted = f"$-total_food_sales:,.2f""
```

FINAL ANSWER: `$89706.00` **Ground truth:** `89706.00` ✓

Figure 2 To answer GAIA, an AI assistant such as GPT4 (here equipped with a code interpreter) needs to complete a few steps, potentially using tools or reading files.

GAIA: A Benchmark for General AI Assistants [Mialon et. al. 2023]

Web browsing

	Allrecipes	Amazon	Apple	ArXiv	GitHub	Booking	ESPN	Coursera
GPT-4 (All Tools)	11.1%	17.1%	44.2%	14.0%	48.8%	22.7%	31.8%	31.0%
$\text{WebVoyager}_{\text{Text-only}}$	**55.6%**	31.7%	34.9%	32.6%	61.0%	2.3%	36.4%	23.8%
WebVoyager	53.3%	**58.5%**	**65.1%**	**51.2%**	**63.4%**	**43.2%**	**38.6%**	**73.8%**
$\textit{WebVoyager}_{\textit{Text-only}}{}^*$	57.8%$_{\pm 0.0\%}$	43.1%$_{\pm 1.4\%}$	36.4%$_{\pm 3.5\%}$	50.4%$_{\pm 1.4\%}$	63.4%$_{\pm 2.5\%}$	2.3%$_{\pm 0.0\%}$	38.6%$_{\pm 2.3\%}$	24.6%$_{\pm 1.4\%}$
*WebVoyager**	51.1%$_{\pm 2.2\%}$	52.9%$_{\pm 1.4\%}$	62.8%$_{\pm 2.3\%}$	52.0%$_{\pm 1.3\%}$	59.3%$_{\pm 3.7\%}$	32.6%$_{\pm 2.7\%}$	47.0%$_{\pm 1.3\%}$	57.9%$_{\pm 2.7\%}$
$\textit{WebVoyager}_{\textit{Claude}}{}^*$	45.9%$_{\pm 3.4\%}$	58.6%$_{\pm 4.2\%}$	58.1%$_{\pm 4.0\%}$	55.0%$_{\pm 7.0\%}$	56.9%$_{\pm 1.4\%}$	19.0%$_{\pm 1.3\%}$	46.2%$_{\pm 1.3\%}$	68.2%$_{\pm 1.3\%}$
$\textit{WebVoyager}_{\textit{GPT-4o}}{}^*$	56.3%$_{\pm 1.3\%}$	53.7%$_{\pm 2.5\%}$	56.6%$_{\pm 1.3\%}$	60.5%$_{\pm 0.0\%}$	57.7%$_{\pm 3.7\%}$	43.9%$_{\pm 3.5\%}$	44.0%$_{\pm 2.7\%}$	65.1%$_{\pm 2.8\%}$

	Cambridge Dictionary	BBC News	Google Flights	Google Map	Google Search	Huggingface	Wolfram Alpha	Overall
GPT-4 (All Tools)	25.6%	9.5%	2.4%	53.7%	60.5%	37.2%	52.2%	30.8%
$\text{WebVoyager}_{\text{Text-only}}$	62.8%	45.2%	7.1%	61.0%	67.4%	20.9%	58.7%	40.1%
WebVoyager	**65.1%**	**61.9%**	**59.5%**	**70.7%**	**76.7%**	**44.2%**	**63.0%**	**59.1%**
$\textit{WebVoyager}_{\textit{Text-only}}{}^*$	66.7%$_{\pm 3.6\%}$	45.2%$_{\pm 2.4\%}$	7.1%$_{\pm 0.0\%}$	62.6%$_{\pm 2.8\%}$	75.2%$_{\pm 1.3\%}$	31.0%$_{\pm 1.4\%}$	60.2%$_{\pm 1.3\%}$	44.3%$_{\pm 0.6\%}$
*WebVoyager**	71.3%$_{\pm 1.3\%}$	60.3%$_{\pm 2.8\%}$	51.6%$_{\pm 1.4\%}$	64.3%$_{\pm 2.8\%}$	77.5%$_{\pm 2.7\%}$	55.8%$_{\pm 2.3\%}$	60.9%$_{\pm 2.2\%}$	57.1%$_{\pm 0.2\%}$
$\textit{WebVoyager}_{\textit{Claude}}{}^*$	71.3%$_{\pm 3.6\%}$	66.7%$_{\pm 4.8\%}$	15.1%$_{\pm 5.5\%}$	55.3%$_{\pm 1.4\%}$	72.9%$_{\pm 1.3\%}$	53.5%$_{\pm 4.7\%}$	51.5%$_{\pm 5.4\%}$	52.8%$_{\pm 1.4\%}$
$\textit{WebVoyager}_{\textit{GPT-4o}}{}^*$	82.2%$_{\pm 1.3\%}$	54.8%$_{\pm 2.4\%}$	28.6%$_{\pm 0.0\%}$	56.9%$_{\pm 2.8\%}$	63.6%$_{\pm 1.3\%}$	42.6%$_{\pm 3.6\%}$	65.2%$_{\pm 2.2\%}$	55.5%$_{\pm 0.8\%}$

Table 1: The main result for WebVoyager. Each website contains 40 to 45 tasks, and we report the Task Success Rate in the table. We show the results of GPT-4 (All Tools), $\text{WebVoyager}_{\text{Text-only}}$ using the accessibility tree, and WebVoyager by comparing with human expert labels. *WebVoyager**, $\textit{WebVoyager}_{\textit{Text-only}}{}^*$, $\textit{WebVoyager}_{\textit{Claude}}{}^*$ and $\textit{WebVoyager}_{\textit{GPT-4o}}{}^*$ are results evaluated by GPT-4V (full trajectory, kappa = 0.70). For each automatic evaluation, we run GPT-4V evaluator three times to calculate the performance mean and standard deviation.

WebVoyager : Building an End-to-End Web Agent with Large Multimodal Models [He et. al. 2024]

Web browsing

Main reasons for Failure	Ratio
Navigation Stuck	44.4%
Visual Grounding Issue	24.8%
Hallucination	21.8%
Prompt Misalignment	9.0%

Table 4: Distribution of main failure reasons.

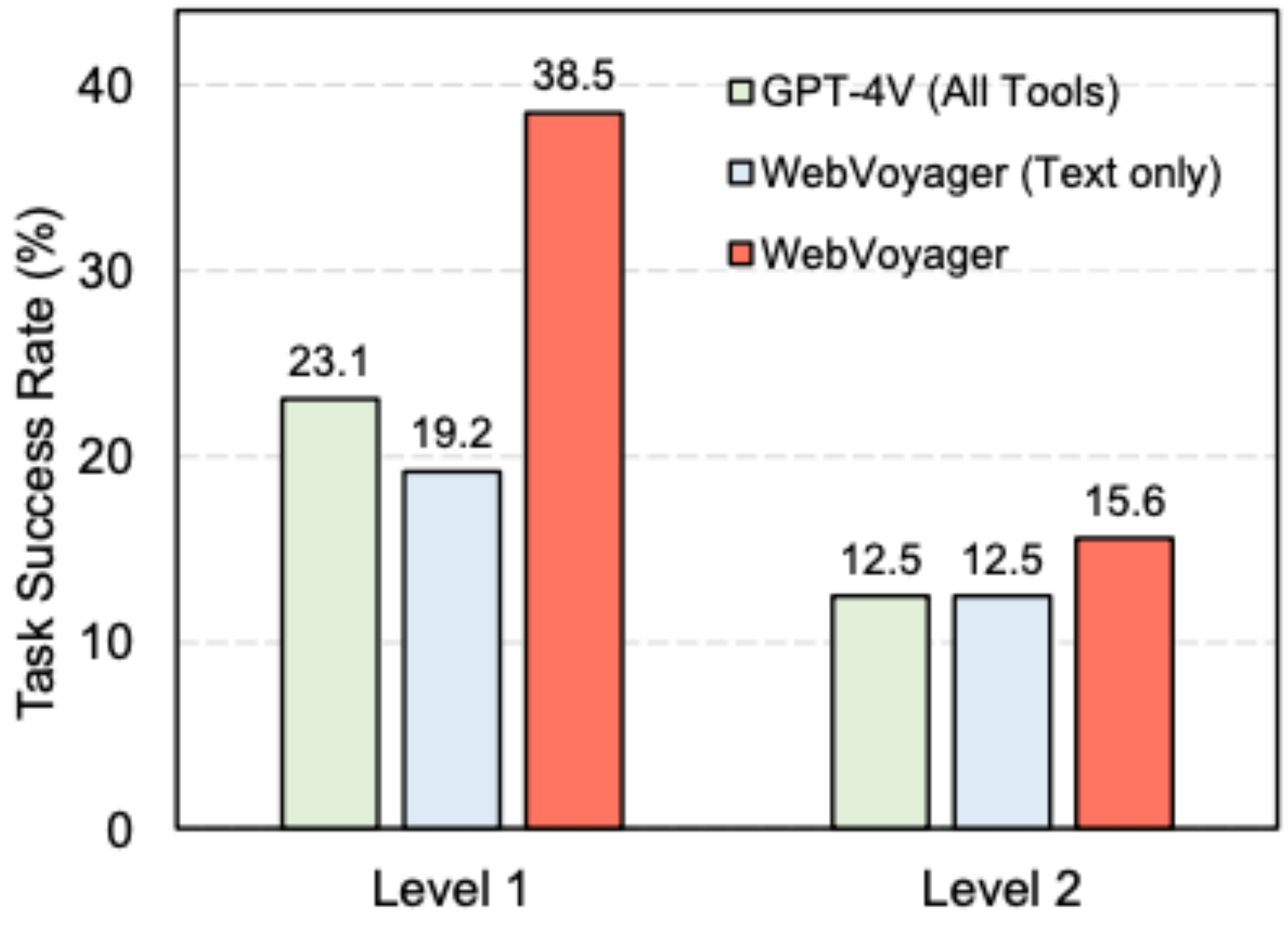


Figure 5: Success Rate for GAIA Level 1 and Level 2.

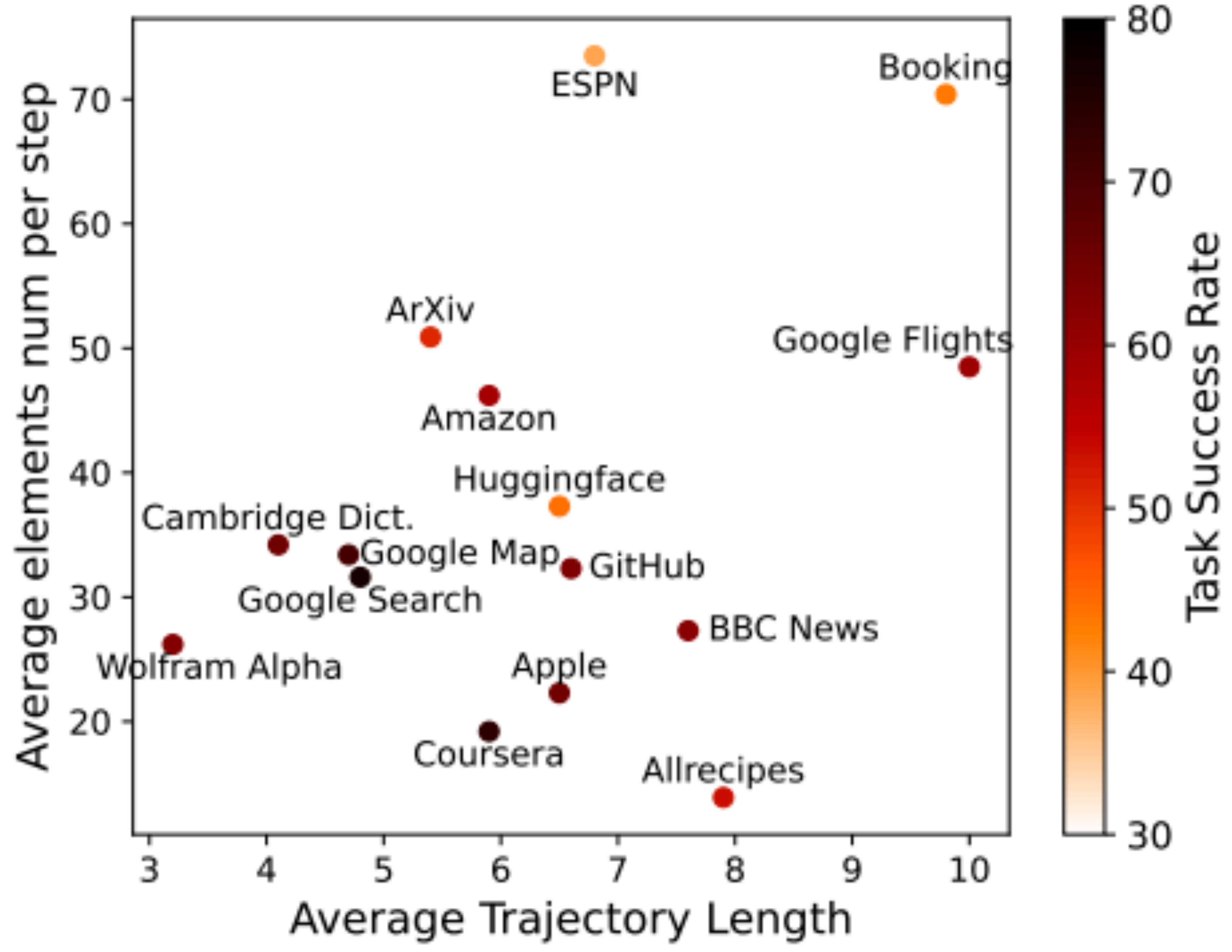
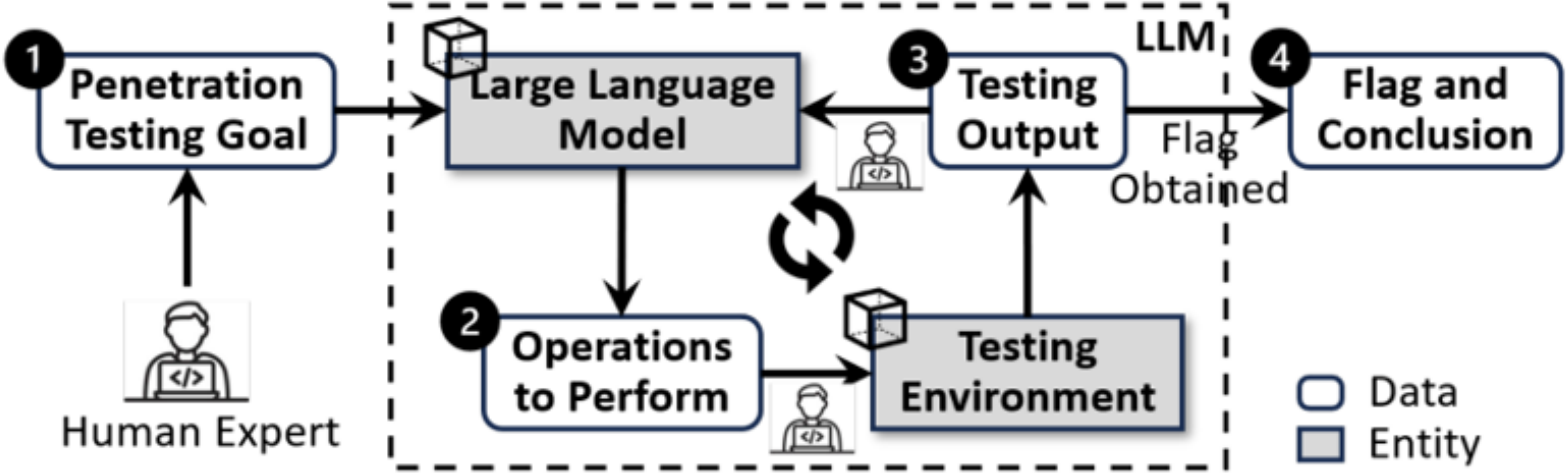


Figure 6: Factors related to task success rate. We show the average number of elements per page and the average trajectory length for each website type. The darker colors indicate a higher task success rate.

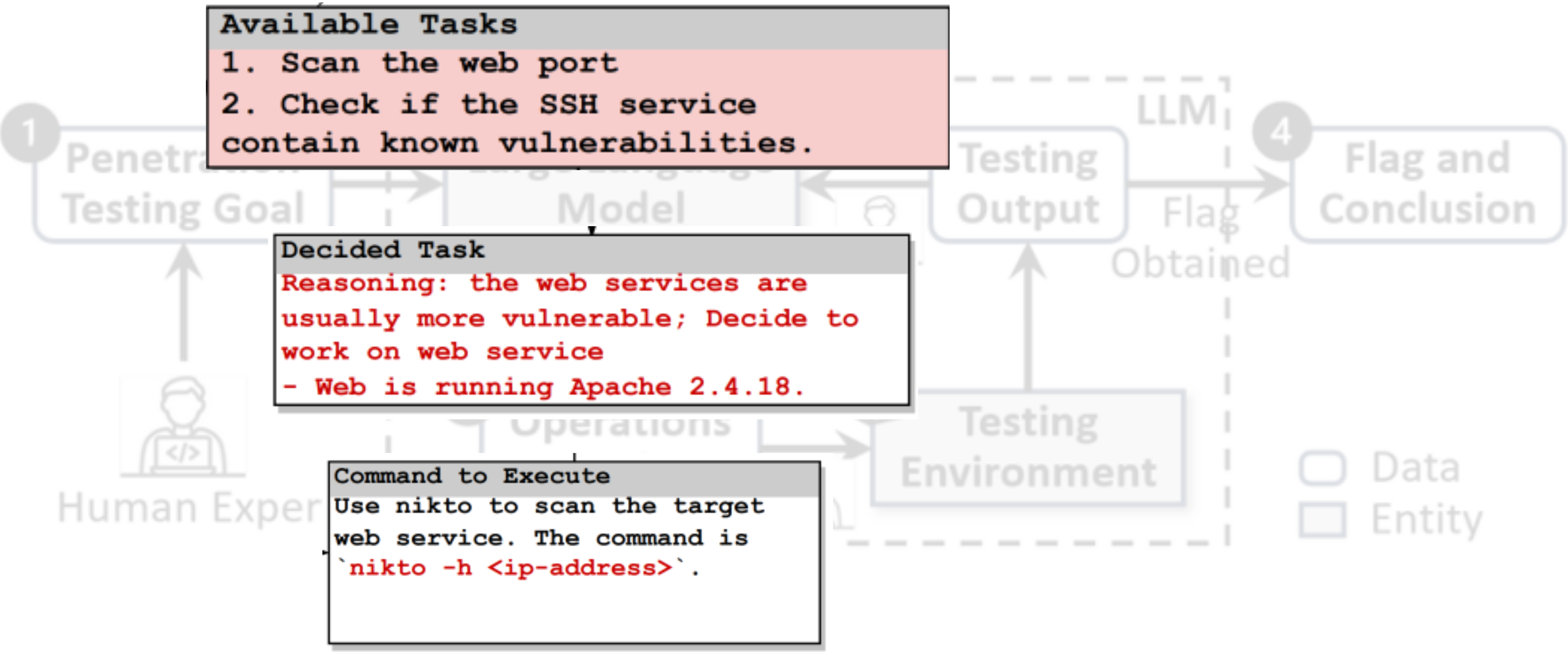
WebVoyager : Building an End-to-End Web Agent with Large Multimodal Models [He et. al. 2024]

Vulnerability detection



PENTESTGPT: An LLM-empowered Automatic Penetration Testing Tool [Deng et. al. 2023]

Vulnerability detection



PENTESTGPT: An LLM-empowered Automatic Penetration Testing Tool [Deng et. al. 2023]

Software engineering

https://github.com/SWE-agent/SWE-agent

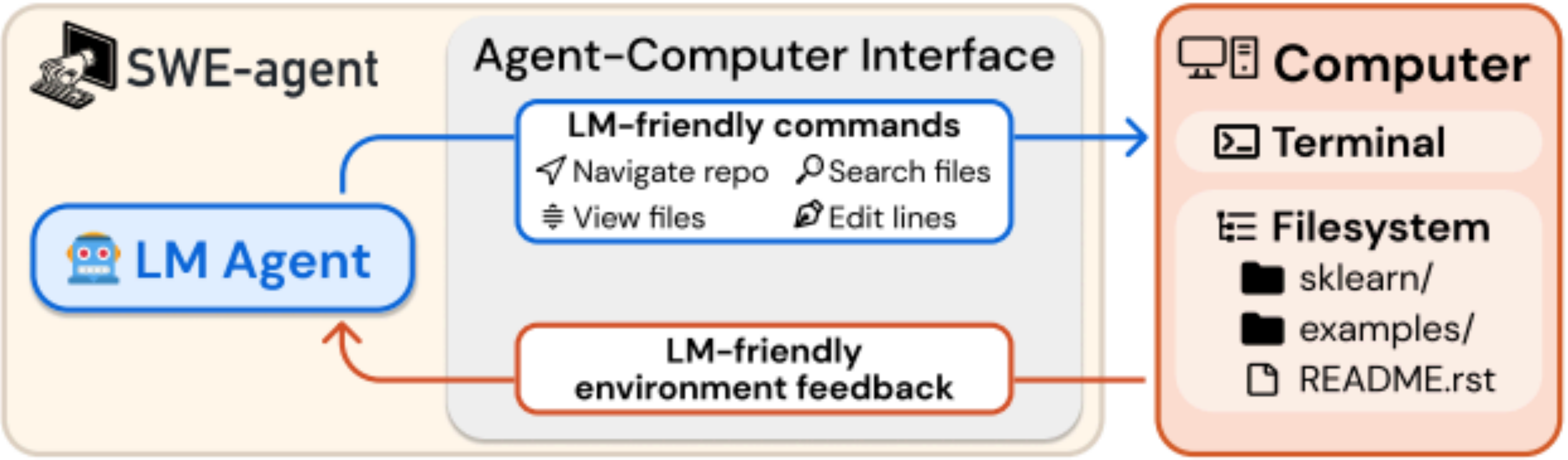


Figure 1: SWE-agent is an LM interacting with a computer through an agent-computer interface (ACI), which includes the commands the agent uses and the format of the feedback from the computer.

SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering [Yang et. al. 2024]

Software engineering

https://github.com/SWE-agent/SWE-agent

Observation (File Viewer)

```
[File: solvers/diophantine.py (3093 lines total)]
(400 lines above)
401:    elif total_degree == 3:
402:        diop_type = "cubic_thue"
403:
404:    elif (total_degree > 3):
405:        if all(coeff[k] == 1):
406:            diop_type = 'general_sum_even'
407:
408:    if diop_type is not None:
409:        return var, coeff, diop_type
(2684 lines below)
```

(a) Observation from the file viewer.

Response (Thought + Edit Action)

We need to add a check to ensure that all powers are the same. Let's edit the code to include this check.

```
edit 404:407
    elif (total_degree > 3 and
            len(set(k.exp[k.is_Pow])) == 1):
        if all(coeff[k] == 1):
            diop_type = 'general_sum_even'
end_of_edit
```

(b) Action using the edit interface.

Figure 3: The file viewer and edit command are integrated. (a) The file viewer shows the agent the open file's content with line numbers. (b) The agent invokes the edit function to replace lines 404-407 in the open file. After the edit, the file viewer shows the agent the now updated version of the file.

SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering [Yang et. al. 2024]

Software engineering

https://github.com/All-Hands-AI/OpenHands

- Docker sandbox
- API server to execute actions
  - Bash shell
  - Jupyter Python server
  - Chromium browser
- Extensible
  - Agent skills: add tools
  - Agent hub: community contributed agents

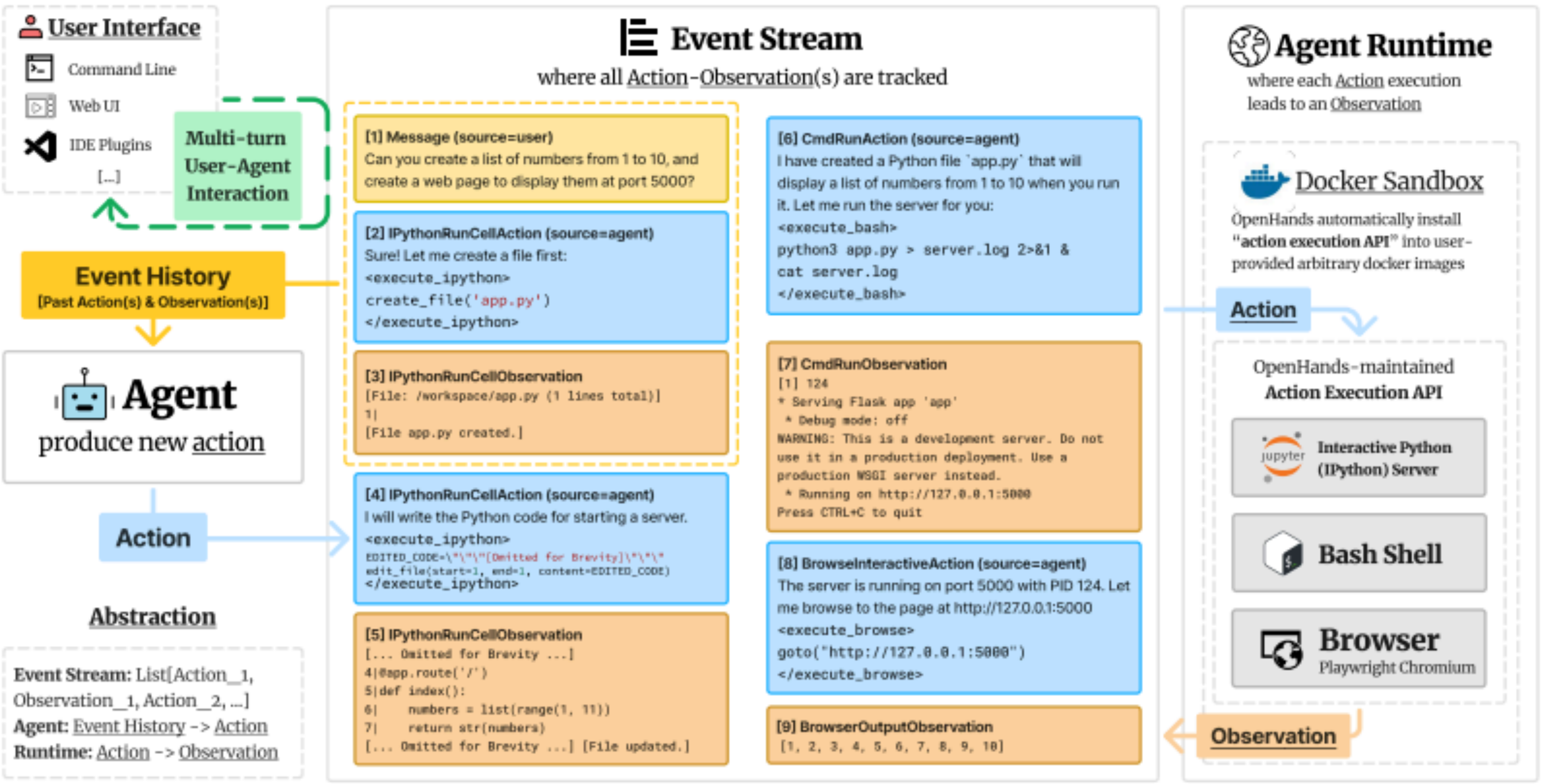


Figure 2: OpenHands consists of 3 main components: 1) **Agent abstraction** where community can contribute different implementation of agents (§2.1) into agenthub (§3); 2) **Event stream** for tracking history of actions and observations; 3) **Runtime** to execute all actions into observations (§2.2).

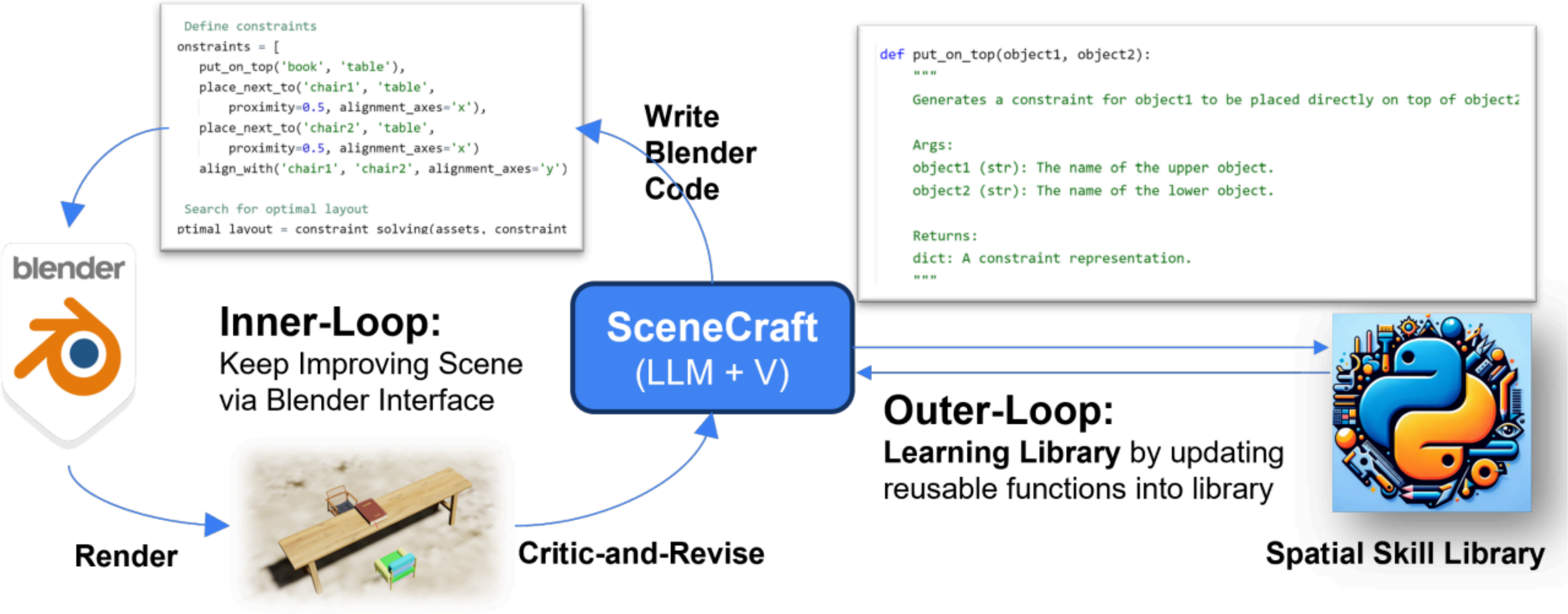
OpenHands: An Open Platform for AI Software Developers as Generalist Agents [Wang et. al. 2024]

Development Agents

- For coding (e.g. SWE-Agent, Aider)
- For broader development (e.g. Devin, OpenHands)

YouTube lecture by Graham Neubig

Scene Generation



SceneCraft: An LLM Agent for Synthesizing 3D Scene as Blender Code [Hu et al, 2024]

Scene Generation

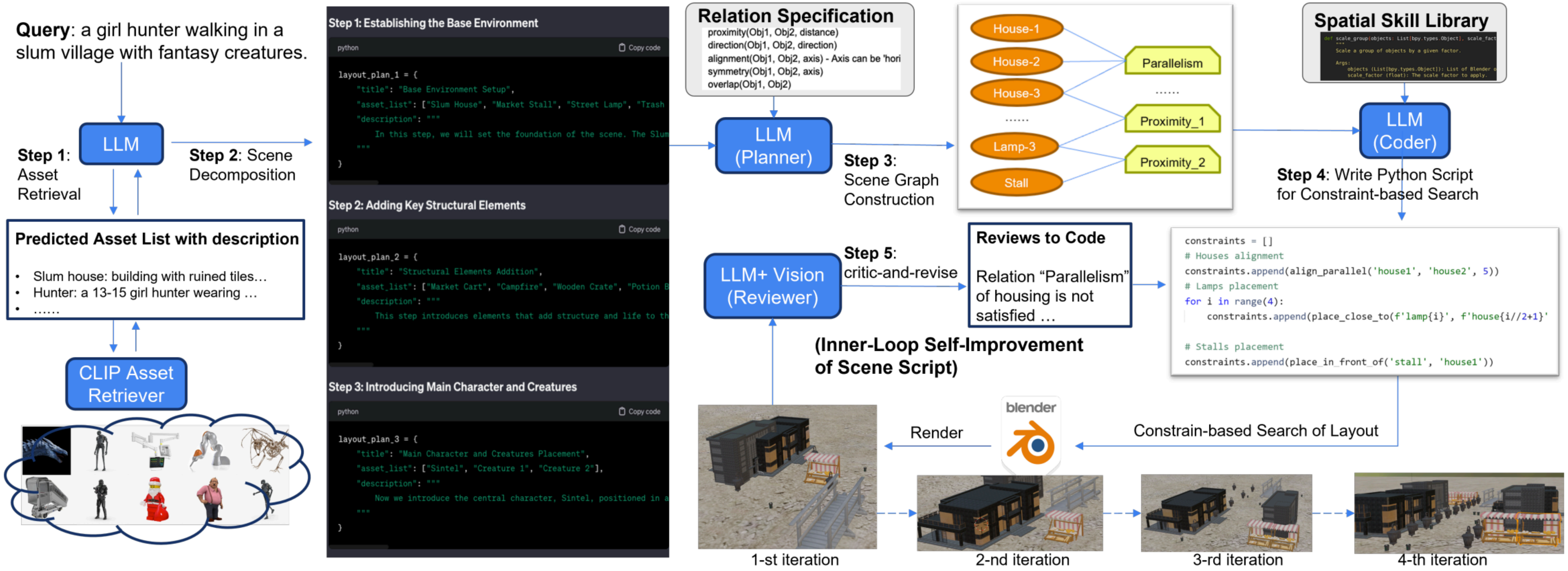
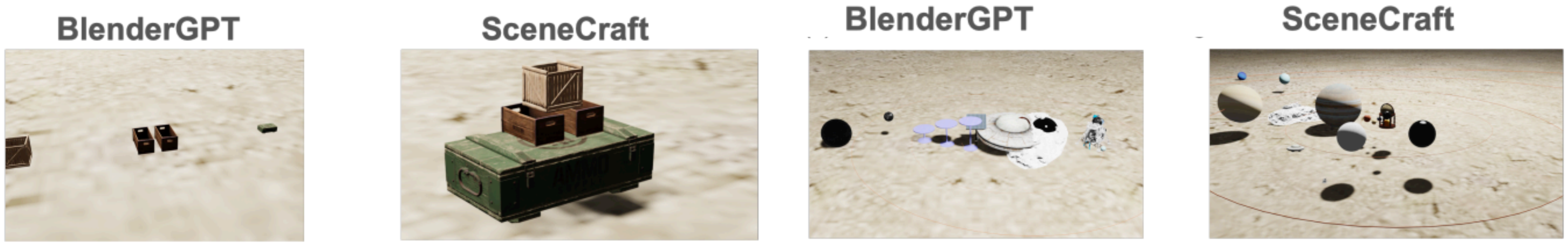


Figure 3. **The workflow of SceneCraft's inner-loop improvement of each scene.** 1) given query, a LLM writes a list of assets descriptions, then use CLIP retriever to fetch assets; 2) then LLM decomposes the full query into a sequence of sub-scene, each associated with a subset of assets and a text description; 3) a LLM-Planner generate a relational graph linking assets to spatial relationship; 4) Based on the graph, LLM-Coder writes python codes to get a list of numerical constraints, which can be executed to search optimal layout, and render into image using Blender; 5) LLM-Reviewer with vision perception capability criticize the rendered image, and update the script accordingly. This critic-and-revise procedure can be done multiple times to iteratively improve the script and scene.

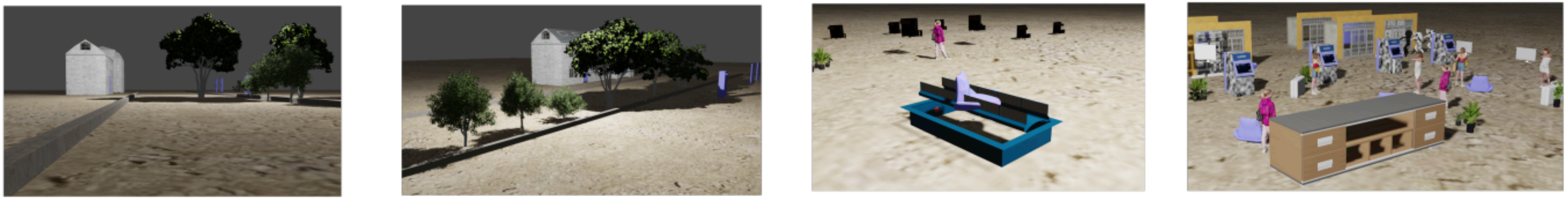
SceneCraft: An LLM Agent for Synthesizing 3D Scene as Blender Code [Hu et al, 2024]

Scene Generation



(a) Three boxes of different sizes, stacked on top of each other

(c) A new solar system with planets orbiting around a small star



(b) Three trees in a row besides a neighborhood

(d) A airport terminal with people, seating areas, and information displays

SceneCraft: An LLM Agent for Synthesizing 3D Scene as Blender Code [Hu et al, 2024]

LLM Agents for Research

Agent Laboratory

https://agentlaboratory.github.io/

- AI agents for ML research
- Collaborate with humans
- Humans focus on ideation and critical thinking
- AI agent help automate repetitive and time-intensive tasks

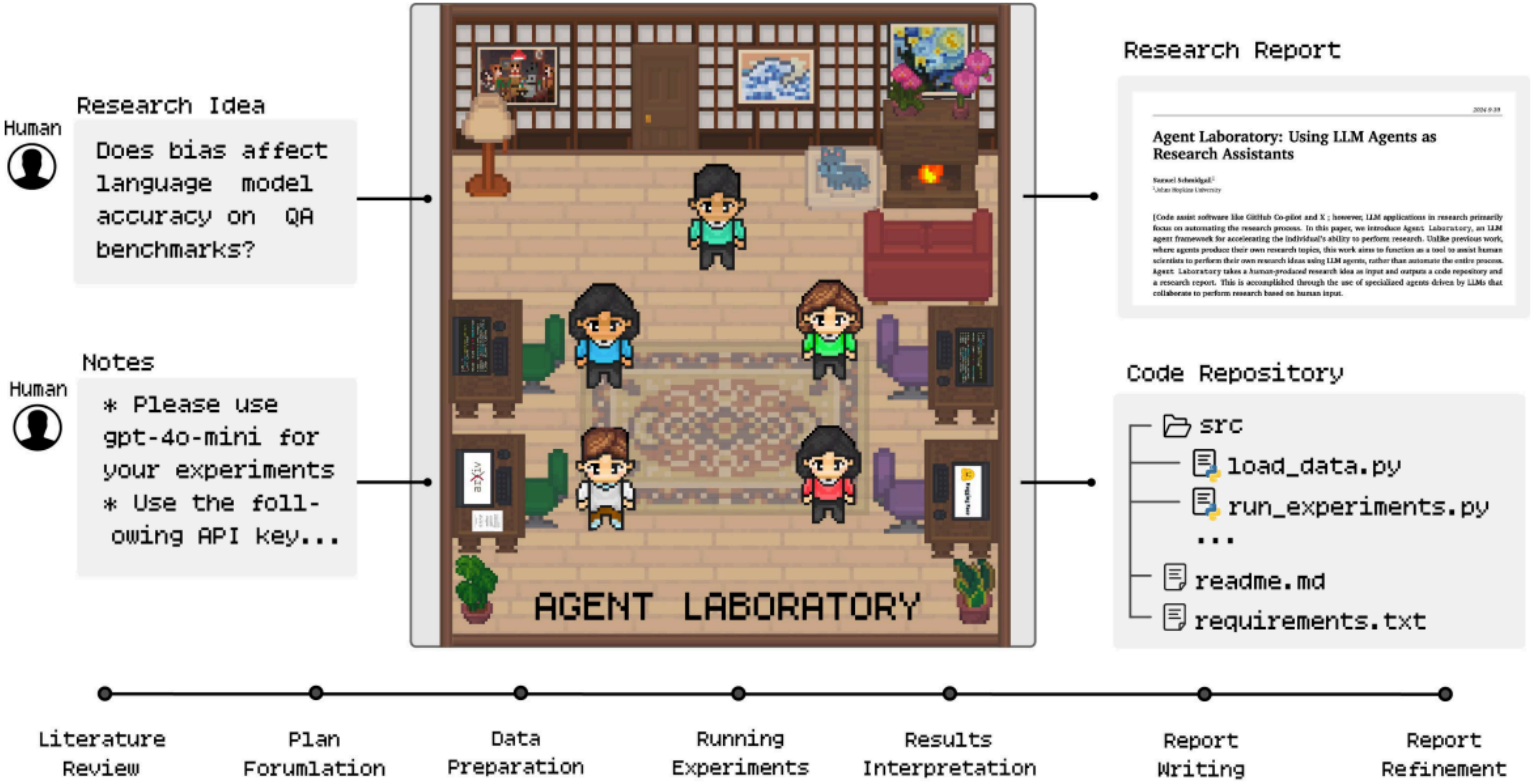


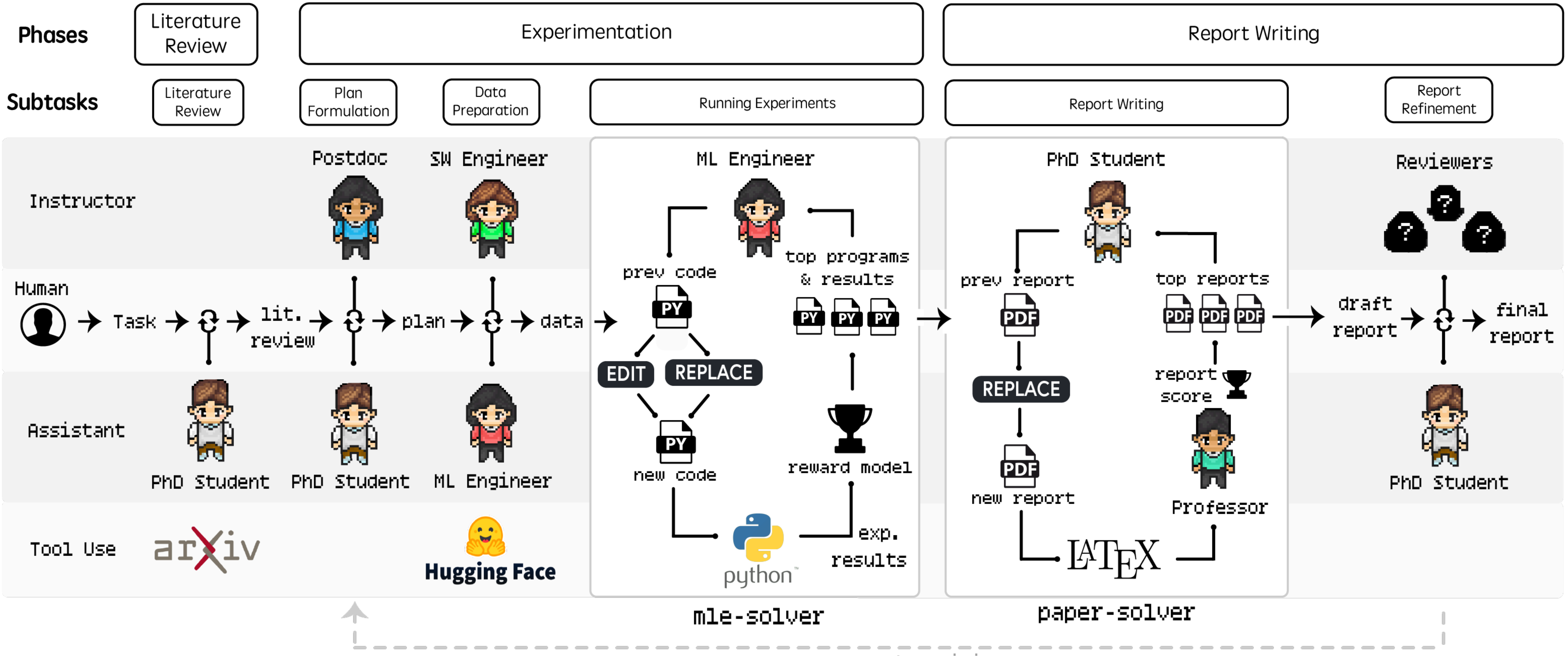
Figure 1 | Agent Laboratory takes as input a human research idea and a set of notes, provides this to a pipeline of specialized LLM-driven agents, and produces a research report and code repository.

Agent Laboratory: Using LLM Agents as Research Assistants [Schmidgall et al, 2025]

Agent Laboratory

https://agentlaboratory.github.io/

- AI agent help with 1) **Literature review**, 2) **Experimentation**, and 3) **Report Writing**



Agent Laboratory: Using LLM Agents as Research Assistants [Schmidgall et al, 2025]

Agent Laboratory

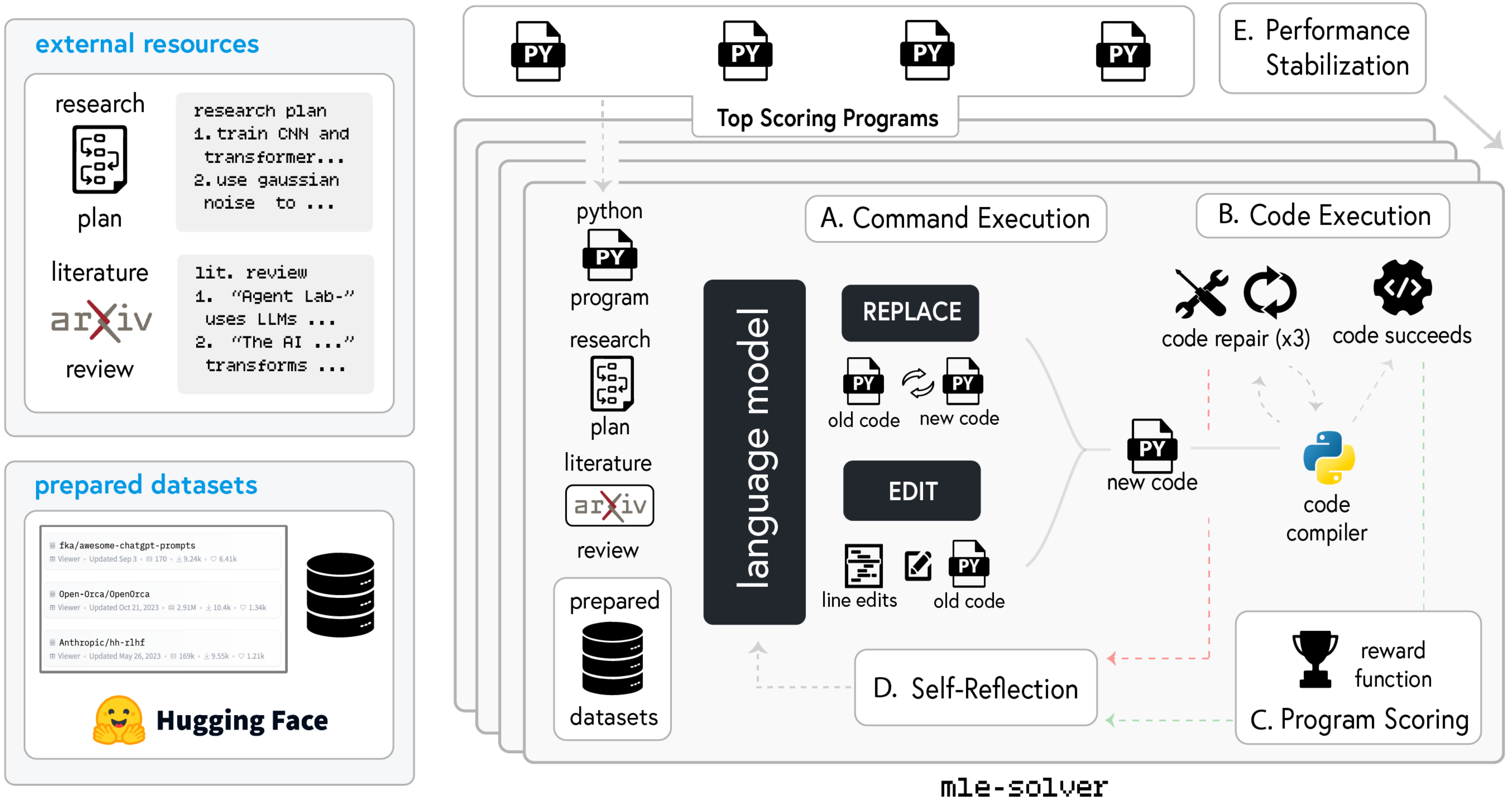
https://agentlaboratory.github.io/

Literature review

- Use arXiv API to summary, full text, add paper

Experiments

- Plan
- Data preparation
- Run experiments
  - mle-solver to write code, test, and refine code
  - Code refined using REPLACE/EDIT
  - Tries to run / fix code up to 3 times
  - Reward model to score effectiveness of ML code
  - Self-Reflection
- Results discussion



[Agent Laboratory: Using LLM Agents as Research Assistants] [Schmidgall et al, 2025]

Agent Laboratory

https://agentlaboratory.github.io/

Evaluation of MLE-Solver on MLE-Bench

Challenge Information			Human Baseline Metrics				MLAB			OpenHands			AIDE (o1-preview)			Agent Laboratory `mle-solver` (ours)		
Challenge Title	Data Type	Min/Max?	Median Score	Bronze Medal	Silver Medal	Gold Medal	Score	Above Median	Medal Earned	Score	Above Median	Medal Earned	Score	Above Median	Medal Earned	Score	Above Median	Medal Earned
detect insults in commentary	💬	Max ⬆	0.778	0.791	0.823	0.833	0.749	❌		0.867	✅	🥇	**0.904**	✅	🥇	**0.839**	✅	🥇
dec 2021 tab playground	▦	Max ⬆	0.953	0.956	0.956	0.956	0.828	❌		0.957	✅	🥇	0.915	❌		**0.961**	✅	🥇
predict trans. conductors	▦	Min ⬇	0.069	0.065	0.062	0.055	0.294	❌		0.183	❌		0.064	✅	🥉	**0.062**	✅	🥈
english text normalization	💬	Max ⬆	0.990	0.990	0.991	0.997	0.0	❌		NR	❌		0.834	❌		**0.990**	✅	🥉
may 2022 tab playground	▦	Max ⬆	0.972	0.998	0.998	0.998	0.711	❌		0.882	❌		0.987	✅		**0.992**	✅	
random acts of pizza	💬	Max ⬆	0.599	0.692	0.724	0.979	0.520	❌		0.591	❌		**0.655**	✅		**0.643**	✅	
spooky author identification	💬	Min ⬇	0.418	0.293	0.269	0.165	0.992	❌		0.582	❌		0.320	✅		**0.532**	❌	
jigsaw toxic comments	💬	Max ⬆	0.980	0.986	0.986	0.987	0.570	❌		0.970	❌		**0.984**	✅		0.874	❌	
russian text normalization	💬	Max ⬆	0.975	0.975	0.982	0.990	0.486	❌		0.486	❌		**0.920**	❌		0.000	❌	
NYC taxi fare prediction	▦	Min ⬇	3.597	2.923	2.881	2.337	1.2e13	❌		355.8	❌		10790	❌		**6.542**	❌	

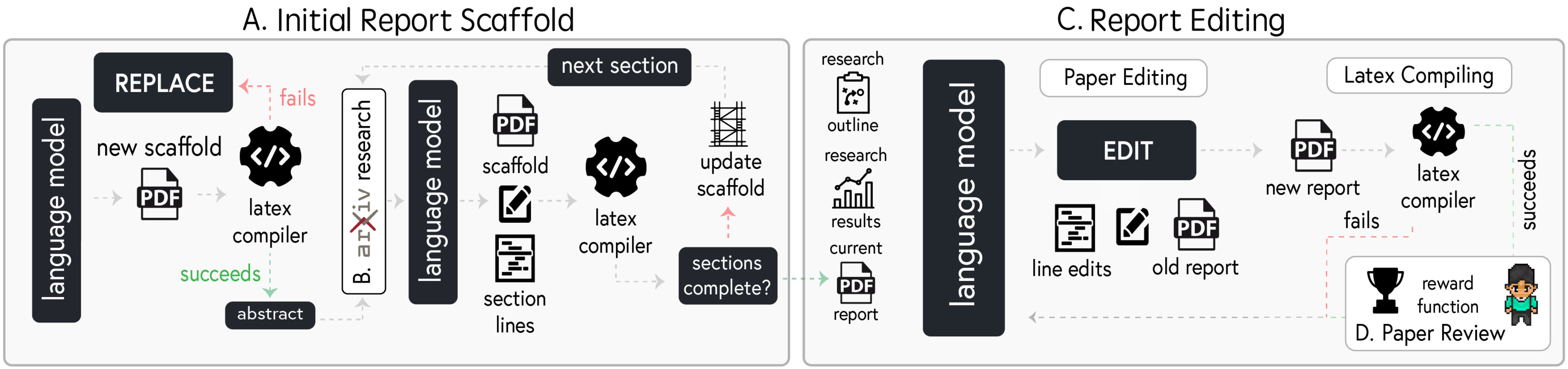
Agent Laboratory: Using LLM Agents as Research Assistants [Schmidgall et al, 2025]

Agent Laboratory

https://agentlaboratory.github.io/

Report writing

- Takes as input research plan, literature review, experimental results, insights and outputs research paper
- Steps: A. Initial report scaffold, B. Arxiv Research, C. Report Editing, D. Paper Review, E. Paper Refinement



paper-solver workflow

Agent Laboratory: Using LLM Agents as Research Assistants [Schmidgall et al, 2025]

Agent Laboratory

Human evaluation of generated papers (15 papers over 5 research questions)

Average human evaluated score by `Agent Laboratory` base LLM

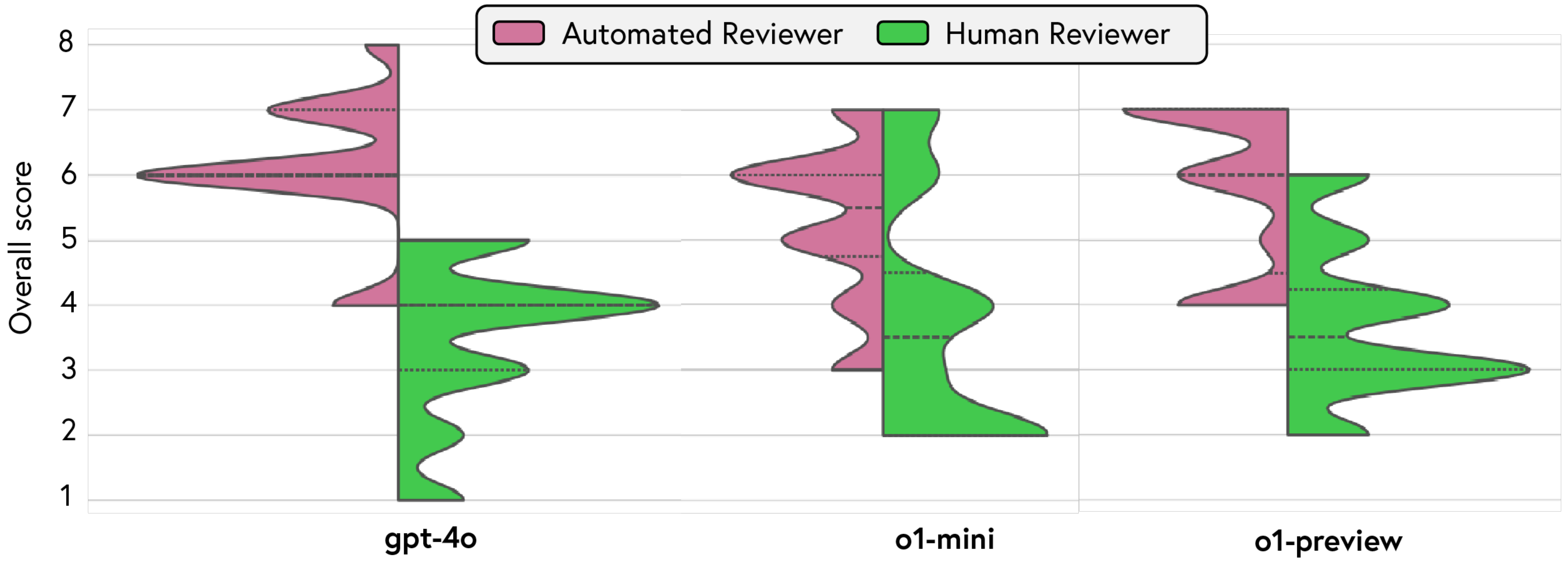
		gpt-4o			o1-mini			o1-preview		
Research Question	Research Type	Experiment Quality	Report Quality	Usefulness	Experiment Quality	Report Quality	Usefulness	Experiment Quality	Report Quality	Usefulness
Are image transformers more or less sensitive to noise than convolutional networks?	Computer Vision	**1.5** / 5	**2.5** / 5	**2.5** / 5	**4.0** / 5	**3.0** / 5	**4.0** / 5	**2.5** / 5	**3.5** / 5	**4.5** / 5
Does gender affect the accuracy on of language models on answering gsm8k questions?	NLP [Social Sci]	**3.0** / 5	**3.0** / 5	**4.0** / 5	**3.0** / 5	**3.5** / 5	**4.0** / 5	**3.0** / 5	**3.5** / 5	**5.0** / 5
Do language models improve accuracy on MedQA when asked to perform differential diagnosis?	NLP [Medical]	**3.0** / 5	**3.5** / 5	**4.5** / 5	**2.5** / 5	**2.5** / 5	**4.5** / 5	**3.5** / 5	**3.5** / 5	**4.0** / 5
Do language models exhibit cognitive biases similar to humans, such as anchoring bias?	NLP [Cog Sci]	**2.5** / 5	**2.5** / 5	**4.5** / 5	**4.0** / 5	**3.5** / 5	**4.5** / 5	**3.0** / 5	**2.0** / 5	**4.0** / 5
Are language models sensitive to word order in multiple choice benchmarks?	NLP [Core]	**3.0** / 5	**3.5** / 5	**4.5** / 5	**2.5** / 5	**3.5** / 5	**4.5** / 5	**2.5** / 5	**4.5** / 5	**4.5** / 5
	Average	**2.6** / 5	**3.0** / 5	**4.0** / 5	**3.2** / 5	**3.2** / 5	**4.3** / 5	**2.9** / 5	**3.4** / 5	**4.4** / 5

Agent Laboratory: Using LLM Agents as Research Assistants [Schmidgall et al, 2025]

Agent Laboratory

Automated evaluation

Comparison of NeurIPS scores (Human Reviewer versus Automated Reviewer) in `Agent Laboratory`



Average Automated Reviewer NeurIPS scores in `Agent Laboratory`

	Quality	Significance	Clarity	Soundness	Presentation	Contribution	Overall
gpt-4o	3.1 / 4	3.1 / 4	3.6 / 4	2.9 / 4	3.4 / 4	3.1 / 4	6.2 / 10
o1-mini	3.1 / 4	2.9 / 4	3.5 / 4	2.9 / 4	3.2 / 4	2.8 / 4	6.0 / 10
o1-preview	3.0 / 4	2.7 / 4	3.7 / 4	3.0 / 4	3.1 / 4	2.7 / 4	5.9 / 10
Average	3.1 / 4	2.9 / 4	3.6 / 4	2.9 / 4	3.2 / 4	2.9 / 4	6.1 / 10

Average Human Reviewer NeurIPS scores in `Agent Laboratory`

	Quality	Significance	Clarity	Soundness	Presentation	Contribution	Overall
gpt-4o	1.8 / 4	2.2 / 4	2.6 / 4	1.8 / 4	2.7 / 4	1.9 / 4	3.5 / 10
o1-mini	2.3 / 4	2.2 / 4	2.1 / 4	1.7 / 4	2.4 / 4	2.2 / 4	3.8 / 10
o1-preview	1.9 / 4	2.5 / 4	2.4 / 4	2.2 / 4	2.4 / 4	2.3 / 4	4.0 / 10
Average	2.0 / 4	2.3 / 4	2.4 / 4	1.9 / 4	2.5 / 4	2.1 / 4	3.8 / 10

Co-Pilot Mode

Quality Evaluation of `Agent Laboratory`

	Utility	Continuation	Satisfaction	Usability	Experiment Quality	Report Quality	Usefulness
Preselected Topics	3.25 / 5	3.5 / 5	3.5 / 5	4.25 / 5	2.5 / 5	2.75 / 5	3.5 / 5
Custom Topics	3.75 / 5	4.0 / 5	3.75 / 5	3.75 / 5	2.25 / 5	3.5 / 5	4.0 / 5
Average	3.5 / 5	3.75 / 5	3.63 / 5	4.0 / 5	2.38 / 5	3.13 / 5	3.75 / 5

Average Self-Evaluation NeurIPS scores in `Agent Laboratory`

	Quality	Significance	Clarity	Soundness	Presentation	Contribution	Overall
Preselected Topics	2.0 / 4	2.0 / 4	2.75 / 4	2.25 / 4	3.0 / 4	2.0 / 4	4.0 / 10
Custom Topics	2.25 / 4	2.0 / 4	3.0 / 4	2.25 / 4	2.75 / 4	2.0 / 4	4.25 / 10
Average	2.13 / 4	2.0 / 4	2.88 / 4	2.25 / 4	2.88 / 4	2.0 / 4	4.13 / 10

Average External Evaluation NeurIPS scores in `Agent Laboratory`

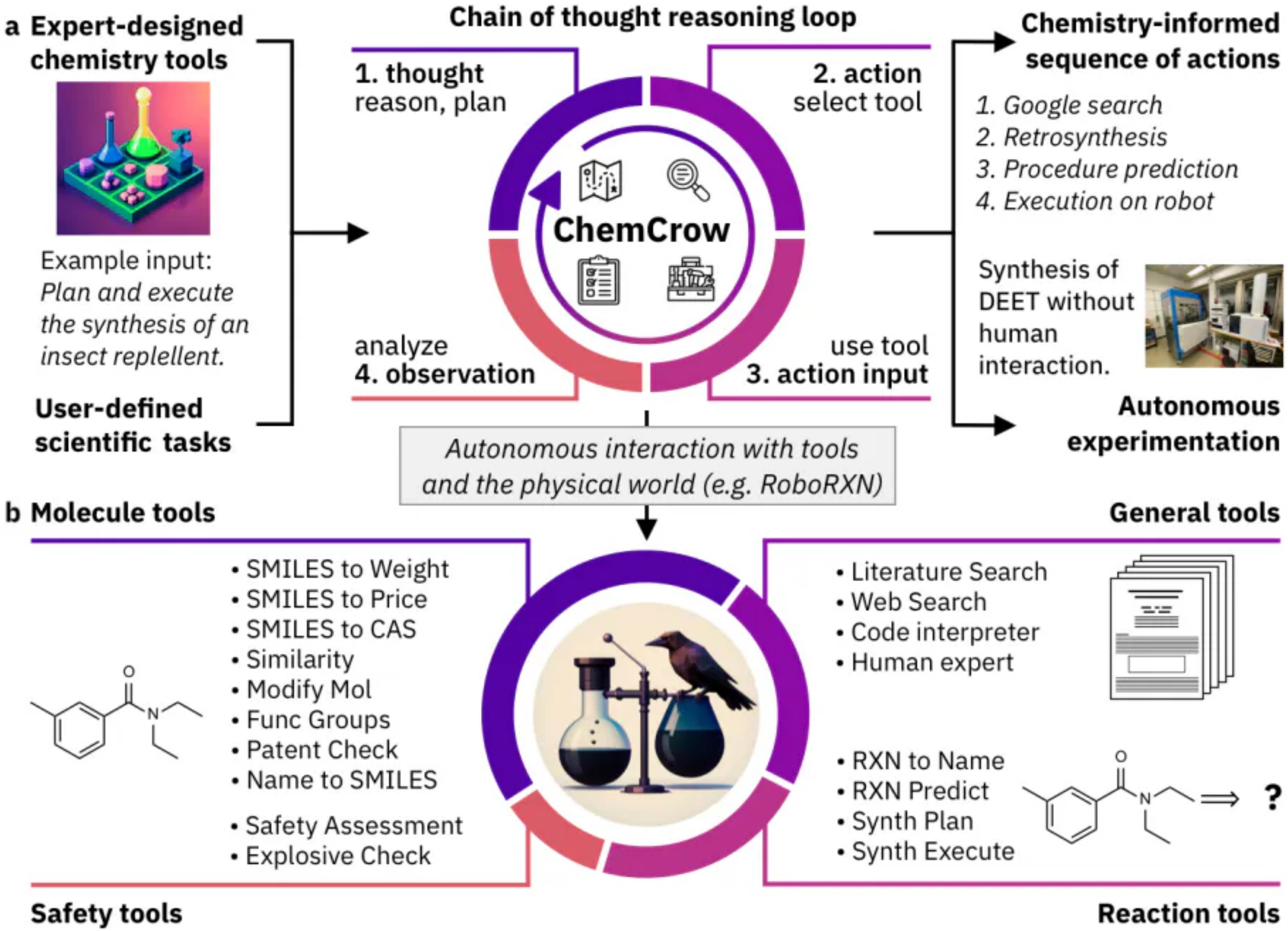
	Quality	Significance	Clarity	Soundness	Presentation	Contribution	Overall
Preselected Topics	3.0 / 4	2.5 / 4	2.75 / 4	2.5 / 4	3.0 / 4	2.0 / 4	4.5 / 10
Custom Topics	2.5 / 4	2.0 / 4	2.5 / 4	2.25 / 4	2.75 / 4	2.25 / 4	4.25 / 10
Average	2.75 / 4	2.25 / 4	2.63 / 4	2.38 / 4	2.88 / 4	2.13 / 4	4.38 / 10
Δ Autonomous	+0.75	-0.05	+0.23	+0.48	+0.33	+0.03	+0.58

Agent Laboratory: Using LLM Agents as Research Assistants [Schmidgall et al, 2025]

ChemCrow: LLM powered chemistry engine

List of tools with name, description, expected input and output

Use ReAct style prompting: thought, action, action input, observation



ChemCrow: Augmenting large-language models with chemistry tools [Bran et al, 2023]

Empowering Biomedical Discovery with AI Agents

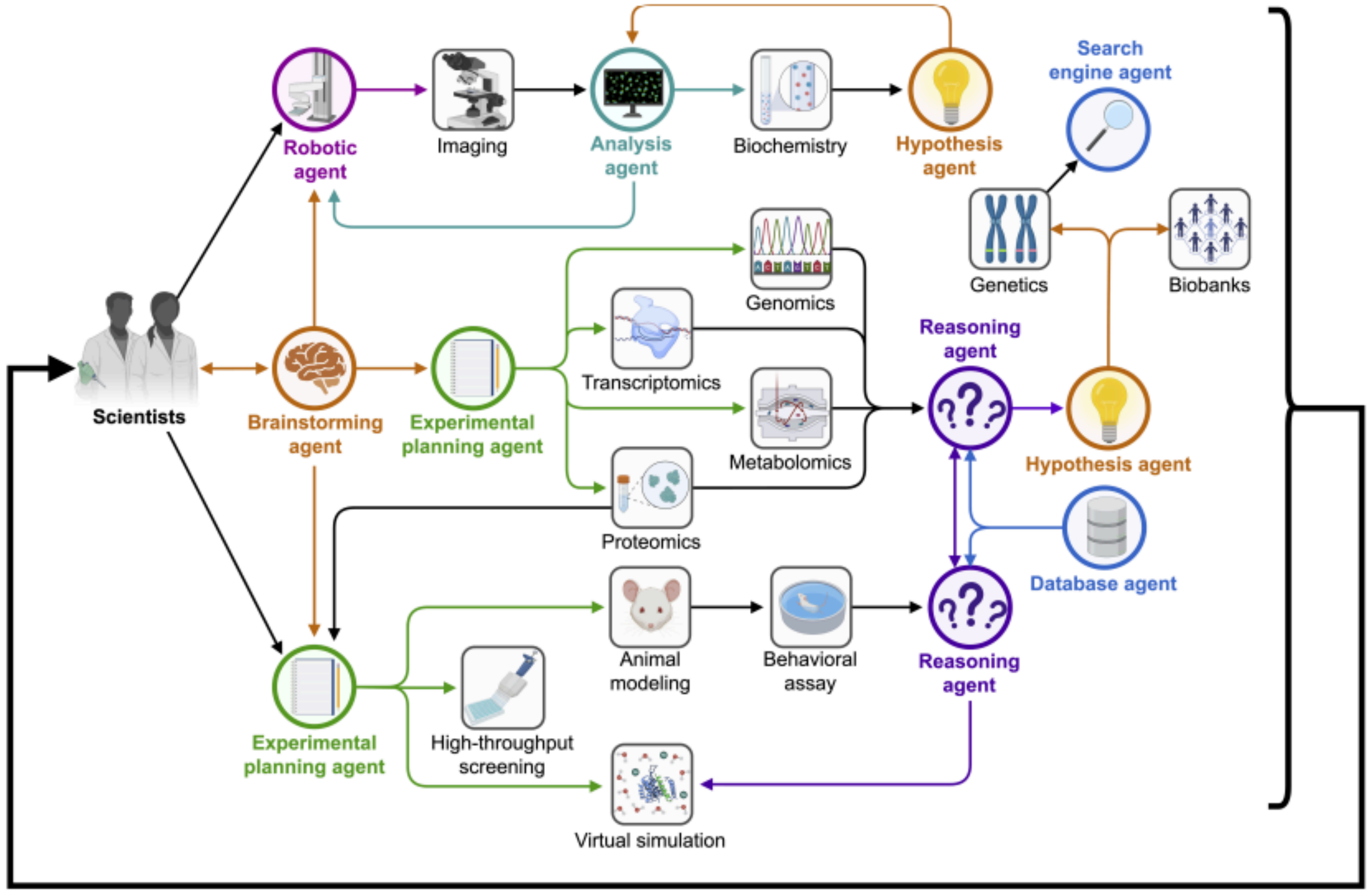


Figure 1: Empowering biomedical research with AI agents. AI agents pave the way for "AI scientists" capable of skeptical learning and reasoning. These multi-agent systems consist of agents based on conversable large language models (LLMs) and can coordinate machine learning (ML) tools, experimental platforms, humans, or even combinations of them. Robotic agent, AI agent that operates robotic hardware for physical experiments; Database agent, AI agent that can information in databases via 'function calling' and APIs; Reasoning agent, AI agent capable of direct reasoning and reasoning with feedback; Hypothesis agent, AI agent that is creative and skeptical when developing hypotheses, capable of characterizing its own uncertainty and using that as a driver to refine its scientific knowledge bases; Brainstorming agent, AI agent that generates a broad spectrum of research ideas; Search engine agent, AI agent that uses search engines as tools to rapidly gather information; Analysis agent, AI agent capable of analyzing experimental results to summarize findings and synthesize concepts; Experimental planning agent, AI agent that optimizes an experimental protocol for execution.

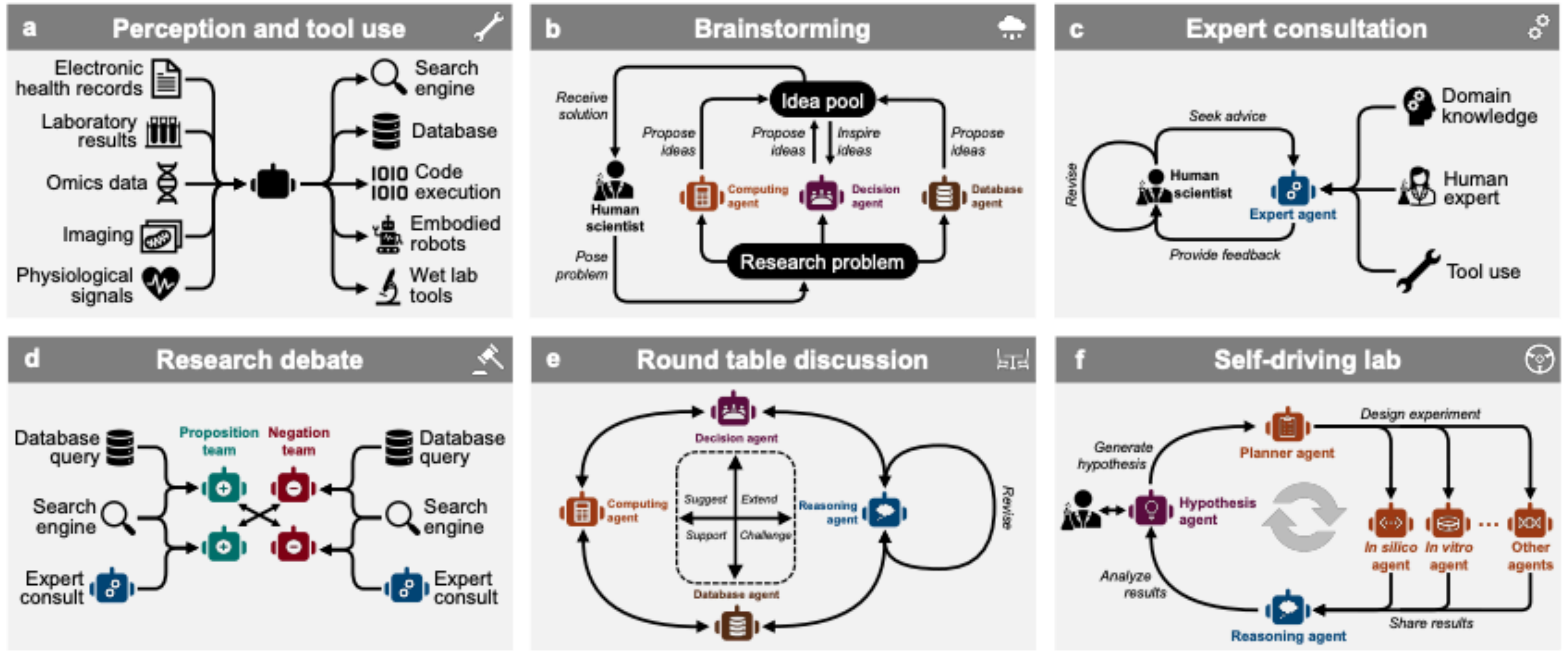


Figure 3: Diverse configurations of AI agents in biology – from an LLM-based AI agent to a multi-agent system with AI models, tools, and integrated physical devices. a. By programming an LLM with the role, one LLM-based agent, equipped with memory and reasoning abilities, performs multi-modal perception and utilizes a range of tools, e.g., web lab tools, to accomplish specified tasks. **b-e.** Leveraging AI agents equipped with diverse roles, perception modules, tools, and domain knowledge enables collaboration between agents and scientists. This collaboration can adopt various schemes, such as expert consultation, debate, brainstorming, and roundtable discussions. **f.** Multi-agent systems can establish a self-driving laboratory wherein numerous agents collaborate on multiple iterations of biological research assisted by humans. Each cycle of research encompasses the generation of hypotheses, the design of experiments, the execution of experiments both in silico and in vitro, and the analysis of results. Computing agent, AI agent that utilizes computational models as tools; Decision agent, AI agent that makes decisions in response to given conditions; Database agent, AI agent that retrieves relevant information from databases; Reasoning agent, AI agent capable of direct reasoning and reasoning with feedback; Expert agent, AI agent that provides professional consultation based on reliable sources, such as domain expertise, feedback from human experts, and the results of specific tools. Hypothesis agent, AI agent capable of skeptical learning and reasoning to generate hypotheses; Planner agent, AI agent that devises plans for future actions; In silico/vitro agent, AI agent that uses tools in silico or in vitro environment.

Empowering Biomedical Discovery with AI Agents [Gao et al, 2024]

Empowering Biomedical Discovery with AI Agents

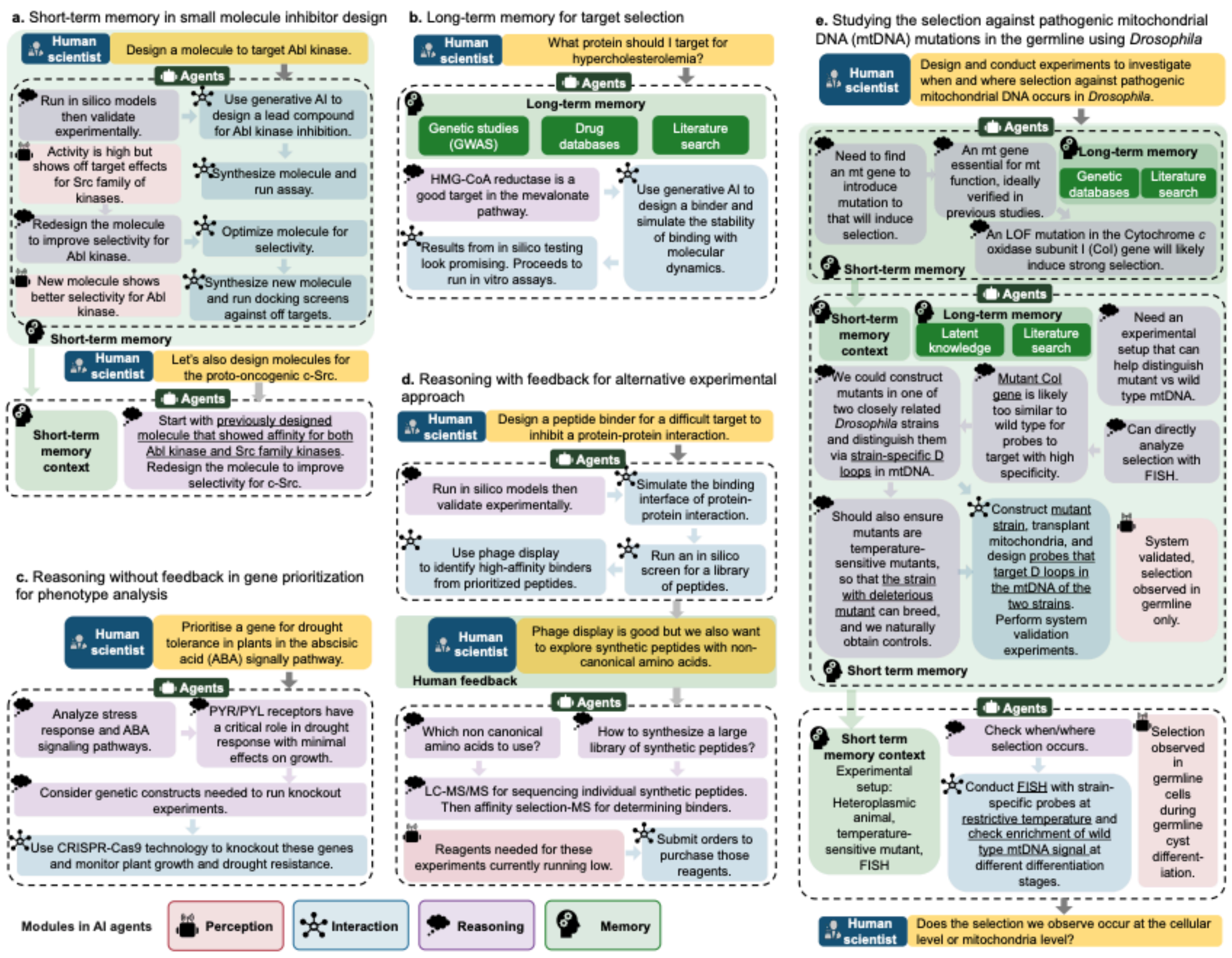


Figure 5: Components of biomedical AI agents. a. Use of a short-term memory module to recall previous relevant experiments for small molecule inhibitor design. **b.** Use of a long-term memory module to retrieve relevant information for target selection for a disease. **c.** Use of reasoning without scientist feedback in gene prioritization for phenotype analysis. **d.** Use of reasoning with feedback from scientists to select an alternative experimental approach.

Empowering Biomedical Discovery with AI Agents [Gao et al, 2024]